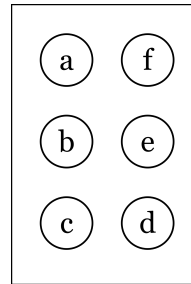


Functions of Two Variables, group *Alpha* α -i

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{4x^2y + xy^2}$, $\frac{\partial f}{\partial y}$... ??? ... $\frac{4x^2+2xy}{2\sqrt{4x^2y+xy^2}}$
 (b) $A = [2, 4]$??? $\left(\frac{80}{2\sqrt{96}}, \frac{32}{2\sqrt{96}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^4, 1, ?]$??? $\frac{x}{e^5} - y - z + 5 = 0$
 (d) $\Delta x = 0.02$, $\Delta y = 0.3$, $D_0 = [e^4, 1]$... ??? ... 4.3
 (e) $h(x, y) = 3x^2 + 2y^2 + xy + 3x$??? $-17/23$
 (f) $y = x - 8$??? $\left[\frac{37}{12}, \frac{-59}{8}\right]$

1.



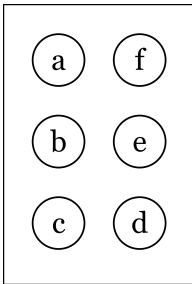
Braille letter

Functions of Two Variables, group *Alpha* α -ii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[4]{x^2y + 2xy^2}$, $\frac{\partial f}{\partial y}$.. ??? .. $\frac{x^2+4xy}{4\sqrt[4]{(x^2y+2xy^2)^3}}$
 (b) $A = [4, 2]$??? $\left(\frac{24}{4\sqrt[4]{64^3}}, \frac{96}{4\sqrt[4]{64^3}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^2, 1, ?]$??? $\frac{x}{e^2} + y + z + 2 = 0$
 (d) $\Delta x = 0.03$, $\Delta y = -0.03$, $D_0 = [e^2, 1]$??? 1.974
 (e) $h(x, y) = 3x^2 + 6y^2 + 6xy - 3x$??? $-53/36$
 (f) $y = -3x + 1$??? $\left[-\frac{1}{2}, \frac{-7}{26}\right]$

2.



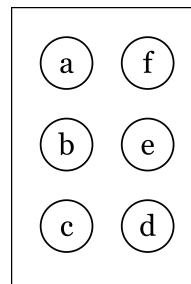
Braille letter

Functions of Two Variables, group *Alpha* α -iii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{2x^2y + xy^2}$, $\frac{\partial f}{\partial x}$.. ??? .. $\frac{4xy+y^2}{2\sqrt{2x^2y+xy^2}}$
 (b) $A = [3, 4]$??? $\left(\frac{64}{2\sqrt{120}}, \frac{42}{2\sqrt{120}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -3, ?]$. ??? . $x + 3z = 0$
 (d) $\Delta x = -0.02$, $\Delta y = -0.3$, $D_0 = [0, -3]$??? 0.1
 (e) $h(x, y) = 7x^2 + 4y^2 + 6xy + 4x$??? $-16/37$
 (f) $y = 4x + 1$??? $\left[\frac{-21}{95}, \frac{11}{47}\right]$

3.



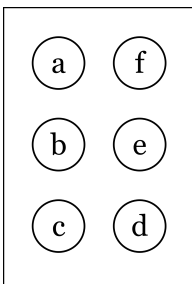
Braille letter

Functions of Two Variables, group *Alpha* α -iv

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[4]{6x^2y + 4xy^2}$, $\frac{\partial f}{\partial y}$. ??? . $\frac{6x^2+8xy}{4\sqrt[4]{(6x^2y+4xy^2)^3}}$
 (b) $A = [2, 3]$??? $\left(\frac{108}{4\sqrt[4]{144^3}}, \frac{72}{4\sqrt[4]{144^3}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^4, 1, ?]$??? $\frac{x}{e^4} - y - z + 4 = 0$
 (d) $\Delta x = 0.2$, $\Delta y = 0.1$, $D_0 = [e^4, 1]$... ??? ... 4.104
 (e) $h(x, y) = -4x^2 - 6y^2 - 5xy + x$??? $6/121$
 (f) $y = -3x + 1$??? $\left[\frac{16}{43}, \frac{-5}{73}\right]$

4.



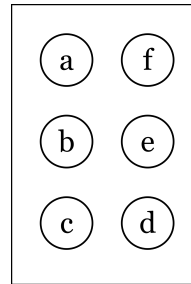
Braille letter

Functions of Two Variables, group *Beta* β -i

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[3]{3x^2y + 5xy^2}$, $\frac{\partial f}{\partial y}$??? $\frac{3x^2+10xy}{3\sqrt[3]{(3x^2y+5xy^2)^2}}$
 (b) $A = [4, 3]$??? $\left(\frac{117}{3\sqrt[3]{324^2}}, \frac{168}{3\sqrt[3]{324^2}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -3, ?]$.. ??? .. $x + 3z = 0$
 (d) $\Delta x = -0.2$, $\Delta y = -0.2$, $D_0 = [0, -3]$??? 0.067
 (e) $h(x, y) = 4x^2 + 3y^2 - xy - 2x$??? $-12/47$
 (f) $y = x + 8$??? $\left[\frac{29}{6}, \frac{29}{6}\right]$

1.



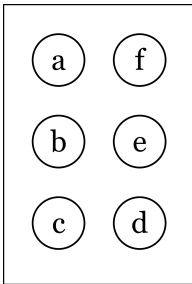
Braille letter

Functions of Two Variables, group *Beta* β -ii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{9x^2y + 5xy^2}$, $\frac{\partial f}{\partial x}$.. ??? .. $\frac{18xy+5y^2}{\sqrt{9x^2y+5xy^2}}$
 (b) $A = [1, 2]$??? $\left(\frac{56}{2\sqrt[3]{38}}, \frac{29}{2\sqrt[3]{38}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^4, 1, ?]$??? $\frac{x}{e^4} + y + z + 4 = 0$
 (d) $\Delta x = -0.3$, $\Delta y = -0.2$, $D_0 = [e^4, 1]$. ??? . 4.295
 (e) $h(x, y) = -x^2 - 7y^2 - xy + 2x$??? $28/29$
 (f) $y = 2x - 2$??? $\left[\frac{30}{31}, \frac{-2}{31}\right]$

2.



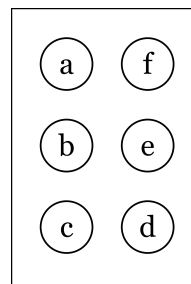
Braille letter

Functions of Two Variables, group *Beta* β -iii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{x^2y + 2xy^2}$, $\frac{\partial f}{\partial y}$... ??? ... $\frac{x^2+4xy}{2\sqrt{x^2y+2xy^2}}$
 (b) $A = [3, 1]$??? $\left(\frac{16}{2\sqrt[3]{15}}, \frac{21}{2\sqrt[3]{15}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [1, 1, ?]$??? $x - y - z + 0 = 0$
 (d) $\Delta x = -0.3$, $\Delta y = -0.02$, $D_0 = [1, 1]$. ??? . 0.0
 (e) $h(x, y) = 3x^2 + 2y^2 + xy + 5x$??? $-50/23$
 (f) $y = 4x - 3$??? $\left[\frac{23}{39}, \frac{-25}{39}\right]$

3.



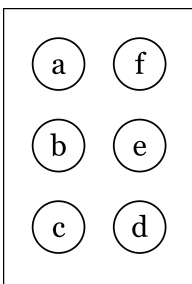
Braille letter

Functions of Two Variables, group *Beta* β -iv

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[3]{x^2y + 3xy^2}$, $\frac{\partial f}{\partial y}$.. ??? .. $\frac{x^2+6xy}{3\sqrt[3]{(x^2y+3xy^2)^2}}$
 (b) $A = [2, 2]$??? $\left(\frac{20}{3\sqrt[3]{32^2}}, \frac{28}{3\sqrt[3]{32^2}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^3, 1, ?]$??? $\frac{x}{e^3} + y + z + 3 = 0$
 (d) $\Delta x = 0.01$, $\Delta y = 0.03$, $D_0 = [e^3, 1]$.. ??? .. 3.03
 (e) $h(x, y) = 2x^2 + 2y^2 - 3xy - 4x$??? $-32/7$
 (f) $y = x + 4$??? $[0, 4]$

4.



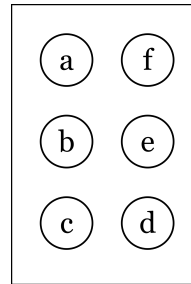
Braille letter

Functions of Two Variables, group *Gamma* γ -i

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{6x^2y + 9xy^2}$, $\frac{\partial f}{\partial y}$. ??? . $\frac{6x^2+18xy}{2\sqrt{6x^2y+9xy^2}}$
 (b) $A = [2, 3]$??? $\left(\frac{153}{2\sqrt{234}}, \frac{132}{2\sqrt{234}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -1, ?]$.. ??? .. $x + z = 1$
 (d) $\Delta x = 0.01$, $\Delta y = -0.02$, $D_0 = [0, -1]$??? 0.02
 (e) $h(x, y) = -2x^2 - 5y^2 + xy - 3x$??? $15/13$
 (f) $y = -2x - 4$??? $\left[\frac{-29}{16}, \frac{-9}{20}\right]$

1.



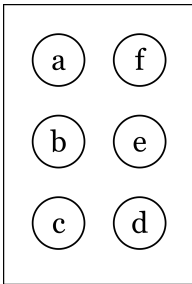
Braille letter

Functions of Two Variables, group *Gamma* γ -ii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{2x^2y + 3xy^2}$, $\frac{\partial f}{\partial x}$??? $\frac{4xy+3y^2}{2\sqrt{2x^2y+3xy^2}}$
 (b) $A = [1, 2]$??? $\left(\frac{20}{2\sqrt{16}}, \frac{28}{2\sqrt{16}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, 1, ?]$.. ??? .. $x - z = 0$
 (d) $\Delta x = -0.03$, $\Delta y = 0.03$, $D_0 = [0, 1]$??? 0.03
 (e) $h(x, y) = x^2 + 5y^2 + 4xy - x$??? $-5/4$
 (f) $y = 2x + 4$??? $\left[\frac{-95}{58}, \frac{21}{13}\right]$

2.



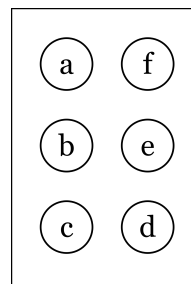
Braille letter

Functions of Two Variables, group *Gamma* γ -iii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[3]{2x^2y + 7xy^2}$, $\frac{\partial f}{\partial x}$??? $\frac{4xy+7y^2}{3\sqrt[3]{(2x^2y+7xy^2)^2}}$
 (b) $A = [4, 3]$??? $\left(\frac{222}{3\sqrt[3]{348^2}}, \frac{200}{3\sqrt[3]{348^2}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -1, ?]$. ??? . $x + z = 0$
 (d) $\Delta x = -0.1$, $\Delta y = 0.3$, $D_0 = [0, -1]$??? -0.3
 (e) $h(x, y) = 3x^2 + 3y^2 - 3xy - 3x$??? $-3/5$
 (f) $y = -2x - 1$??? $\left[\frac{-2}{7}, \frac{-3}{7}\right]$

3.



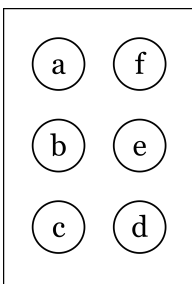
Braille letter

Functions of Two Variables, group *Gamma* γ -iv

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{4x^2y + 3xy^2}$, $\frac{\partial f}{\partial x}$??? $\frac{8xy+3y^2}{2\sqrt{4x^2y+3xy^2}}$
 (b) $A = [4, 2]$??? $\left(\frac{152}{2\sqrt{176}}, \frac{112}{2\sqrt{176}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, 2, ?]$??? $x - 2z = 1$
 (d) $\Delta x = 0.03$, $\Delta y = 0.2$, $D_0 = [0, 2]$. ??? . 0.1
 (e) $h(x, y) = 4x^2 + 5y^2 + 2xy + 3x$.. ??? .. $-45/76$
 (f) $y = -x + 3$??? $\left[\frac{3}{2}, \frac{21}{22}\right]$

4.



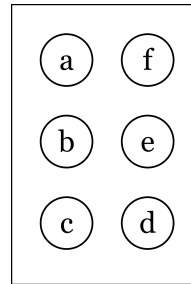
Braille letter

Functions of Two Variables, group *Delta* δ -i

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[4]{3x^2y + 9xy^2}$, $\frac{\partial f}{\partial x}$. ??? . $\frac{6xy+9y^2}{4\sqrt[4]{(3x^2y+9xy^2)^3}}$
 (b) $A = [4, 1]$??? $\left(\frac{33}{4\sqrt[4]{84^3}}, \frac{120}{4\sqrt[4]{84^3}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^4, 1, ?]$??? $\frac{x}{e^4} - y - z + 4 = 0$
 (d) $\Delta x = 0.01$, $\Delta y = -0.3$, $D_0 = [e^4, 1]$.. ??? .. 3.7
 (e) $h(x, y) = -5x^2 - 2y^2 + xy - 2x$??? $3/13$
 (f) $y = 5x + 1$??? $\left[\frac{-21}{100}, \frac{-1}{24}\right]$

1.



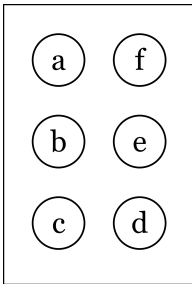
Braille letter

Functions of Two Variables, group *Delta* δ -ii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[3]{2x^2y + 9xy^2}$, $\frac{\partial f}{\partial y}$??? $\frac{2x^2+18xy}{3\sqrt[3]{(2x^2y+9xy^2)^2}}$
 (b) $A = [2, 4]$??? $\left(\frac{176}{3\sqrt[3]{320^2}}, \frac{304}{3\sqrt[3]{320^2}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, 2, ?]$.. ??? .. $x - 2z = 0$
 (d) $\Delta x = 0.3$, $\Delta y = -0.02$, $D_0 = [0, 2]$??? -0.01
 (e) $h(x, y) = -7x^2 - 3y^2 + xy + 3x$??? $27/83$
 (f) $y = x + 5$??? $\left[\frac{-11}{9}, \frac{34}{11}\right]$

2.



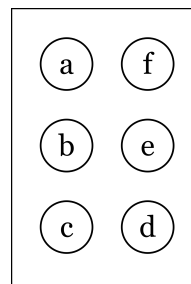
Braille letter

Functions of Two Variables, group *Delta* δ -iii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[3]{2x^2y + xy^2}$, $\frac{\partial f}{\partial y}$. ??? . $\frac{2x^2+2xy}{3\sqrt[3]{(2x^2y+xy^2)^2}}$
 (b) $A = [1, 3]$??? $\left(\frac{21}{3\sqrt[3]{15^2}}, \frac{8}{3\sqrt[3]{15^2}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [1, 1, ?]$??? $x - y - z + 0 = 0$
 (d) $\Delta x = 0.2$, $\Delta y = 0.2$, $D_0 = [1, 1]$... ??? ... 0.0
 (e) $h(x, y) = -7x^2 - 3y^2 + 3xy + 3x$??? $9/31$
 (f) $y = 2x + 8$??? $\left[\frac{-69}{26}, \frac{7}{5}\right]$

3.



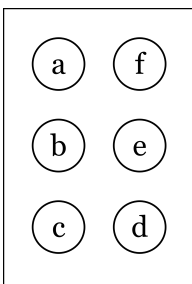
Braille letter

Functions of Two Variables, group *Delta* δ -iv

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[4]{5x^2y + 7xy^2}$, $\frac{\partial f}{\partial y}$??? $\frac{5x^2+14xy}{4\sqrt[4]{(5x^2y+7xy^2)^3}}$
 (b) $A = [2, 1]$??? $\left(\frac{27}{4\sqrt[4]{34^3}}, \frac{96}{4\sqrt[4]{34^3}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, 2, ?]$.. ??? .. $x + 2z = 0$
 (d) $\Delta x = -0.3$, $\Delta y = -0.1$, $D_0 = [0, 2]$??? -0.6
 (e) $h(x, y) = 2x^2 + 2y^2 - xy + 2x$??? $-8/15$
 (f) $y = 3x + 3$??? $\left[\frac{37}{34}, \frac{-3}{34}\right]$

4.



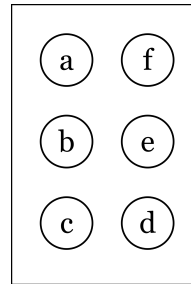
Braille letter

Functions of Two Variables, group *Epsilon* ϵ -i

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[3]{6x^2y + 2xy^2}$, $\frac{\partial f}{\partial x}$??? $\frac{12xy+2y^2}{3\sqrt[3]{(6x^2y+2xy^2)^2}}$
 (b) $A = [4, 4]$??? $\left(\frac{224}{3\sqrt[3]{512^2}}, \frac{160}{3\sqrt[3]{512^2}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, 3, ?]$.. ??? .. $x - 3z = 0$
 (d) $\Delta x = 0.3$, $\Delta y = -0.03$, $D_0 = [0, 3]$. ??? . 0.1
 (e) $h(x, y) = -7x^2 - 4y^2 + 6xy + x$??? $1/19$
 (f) $y = -2x + 6$??? $\left[\frac{19}{10}, 7\right]$

1.



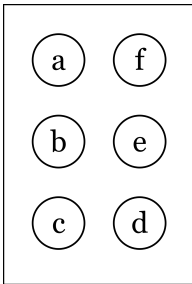
Braille letter

Functions of Two Variables, group *Epsilon* ϵ -ii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{x^2y + 3xy^2}$, $\frac{\partial f}{\partial x}$... ??? ... $\frac{2xy-3y^2}{2\sqrt{x^2y+3xy^2}}$
 (b) $A = [3, 3]$??? $\left(\frac{45}{2\sqrt{108}}, \frac{63}{2\sqrt{108}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^2, 1, ?]$??? $\frac{x}{e^3} - y - z + 3 = 0$
 (d) $\Delta x = -0.3$, $\Delta y = 0.1$, $D_0 = [e^2, 1]$.. ??? .. 2.259
 (e) $h(x, y) = -2x^2 - 4y^2 - 2xy + 4x$??? $16/9$
 (f) $y = 3x - 1$??? $\left[\frac{15}{44}, \frac{1}{44}\right]$

2.



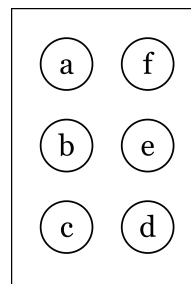
Braille letter

Functions of Two Variables, group *Epsilon* ϵ -iii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[4]{8x^2y + 7xy^2}$, $\frac{\partial f}{\partial x}$??? $\frac{16xy+7y^2}{4\sqrt[4]{(8x^2y+7xy^2)^3}}$
 (b) $A = [2, 2]$??? $\left(\frac{92}{4\sqrt[4]{120^3}}, \frac{176}{4\sqrt[4]{120^3}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -2, ?]$.. ??? .. $x + 2z = 0$
 (d) $\Delta x = -0.01$, $\Delta y = 0.1$, $D_0 = [0, -2]$??? -0.05
 (e) $h(x, y) = 3x^2 + 5y^2 - 4xy + x$??? $-5/44$
 (f) $y = x + 7$??? $\left[\frac{-43}{8}, \frac{13}{8}\right]$

3.



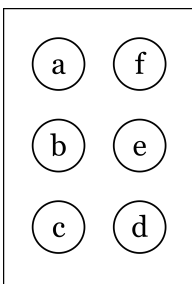
Braille letter

Functions of Two Variables, group *Epsilon* ϵ -iv

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{6x^2y + 8xy^2}$, $\frac{\partial f}{\partial x}$.. ??? .. $\frac{12xy+8y^2}{2\sqrt{6x^2y+8xy^2}}$
 (b) $A = [4, 2]$??? $\left(\frac{256}{2\sqrt{320}}, \frac{224}{2\sqrt{320}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^2, 1, ?]$??? $\frac{x}{e^3} - y - z + 3 = 0$
 (d) $\Delta x = 0.3$, $\Delta y = -0.03$, $D_0 = [e^2, 1]$. ??? . 2.011
 (e) $h(x, y) = 7x^2 + 2y^2 - 4xy - x$??? $-1/20$
 (f) $y = -2x + 7$??? $\left[\frac{85}{46}, \frac{76}{23}\right]$

4.



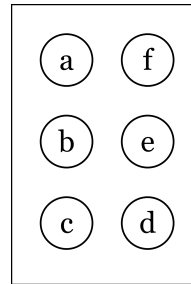
Braille letter

Functions of Two Variables, group *Zeta* ζ -i

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{6x^2y + 5xy^2}$, $\frac{\partial f}{\partial y}$... ??? ... $\frac{6x^2+10xy}{2\sqrt{6x^2y+5xy^2}}$
 (b) $A = [3, 4]$??? $\left(\frac{448}{2\sqrt[4]{456}}, \frac{174}{2\sqrt[4]{456}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [1, 1, ?]$??? $x - y - z + 0 = 0$
 (d) $\Delta x = 0.2$, $\Delta y = -0.2$, $D_0 = [1, 1]$.. ??? .. 0.0
 (e) $h(x, y) = -2x^2 - 2y^2 + xy - 3x$??? $6/5$
 (f) $y = -4x - 1$??? $\left[-\frac{5}{19}, \frac{1}{19}\right]$

1.



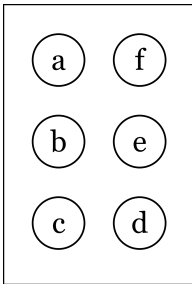
Braille letter

Functions of Two Variables, group *Zeta* ζ -ii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[4]{8x^2y + 7xy^2}$, $\frac{\partial f}{\partial y}$??? $\frac{16x^2+14xy}{4\sqrt[4]{(8x^2y+7xy^2)^3}}$
 (b) $A = [3, 2]$??? $\left(\frac{124}{4\sqrt[4]{228^3}}, \frac{156}{4\sqrt[4]{228^3}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -1, ?]$. ??? . $x + z = 1$
 (d) $\Delta x = 0.1$, $\Delta y = -0.03$, $D_0 = [0, -1]$??? 0.03
 (e) $h(x, y) = -6x^2 - 6y^2 - 3xy - 3x$... ??? ... $11/27$
 (f) $y = 3x + 1$??? $\left[\frac{-7}{23}, \frac{2}{23}\right]$

2.



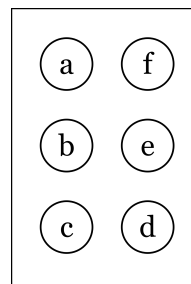
Braille letter

Functions of Two Variables, group *Zeta* ζ -iii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[3]{x^2y + 3xy^2}$, $\frac{\partial f}{\partial x}$??? $\frac{2xy+3y^2}{3\sqrt[3]{(x^2y+3xy^2)^2}}$
 (b) $A = [1, 1]$??? $\left(\frac{5}{3\sqrt[3]{4^2}}, \frac{14}{3\sqrt[3]{4^2}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -1, ?]$. ??? . $x + z = 0$
 (d) $\Delta x = -0.2$, $\Delta y = 0.3$, $D_0 = [0, -1]$??? -0.3
 (e) $h(x, y) = -x^2 - 3y^2 + xy + 3x$??? $27/11$
 (f) $y = -2x + 4$??? $\left[\frac{11}{6}, \frac{1}{3}\right]$

3.



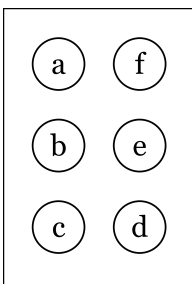
Braille letter

Functions of Two Variables, group *Zeta* ζ -iv

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[4]{5x^2y + 5xy^2}$, $\frac{\partial f}{\partial y}$??? $\frac{5x^2+10xy}{4\sqrt[4]{(5x^2y+5xy^2)^3}}$
 (b) $A = [1, 1]$??? $\left(\frac{30}{4\sqrt[4]{10^3}}, \frac{15}{4\sqrt[4]{10^3}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -2, ?]$. ??? . $x - 2z = 0$
 (d) $\Delta x = 0.01$, $\Delta y = -0.1$, $D_0 = [0, -2]$??? 0.05
 (e) $h(x, y) = -2x^2 - 2y^2 - 2xy + 2x$??? $2/3$
 (f) $y = -x + 2$??? $\left[-\frac{5}{2}, \frac{1}{2}\right]$

4.



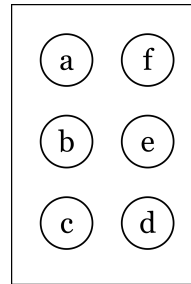
Braille letter

Functions of Two Variables, group *Eta* η -i

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{3x^2y + 5xy^2}$, $\frac{\partial f}{\partial y}$... ??? ... $\frac{3x^2+10xy}{2\sqrt{3x^2y+5xy^2}}$
 (b) $A = [2, 3]$??? $\left(\frac{81}{2\sqrt{126}}, \frac{144}{2\sqrt{126}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^4, 1, ?]$??? $\frac{x}{e^5} - y - z + 5 = 0$
 (d) $\Delta x = -0.02$, $\Delta y = 0.02$, $D_0 = [e^4, 1]$. ??? . 4.02
 (e) $h(x, y) = -6x^2 - 6y^2 - 3xy - 4x$??? $97/135$
 (f) $y = x - 7$??? $\left[\frac{101}{30}, \frac{-109}{18}\right]$

1.



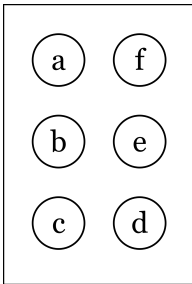
Braille letter

Functions of Two Variables, group *Eta* η -ii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[3]{5x^2y + 3xy^2}$, $\frac{\partial f}{\partial x}$??? $\frac{10xy+3y^2}{3\sqrt[3]{(5x^2y+3xy^2)^2}}$
 (b) $A = [2, 2]$??? $\left(\frac{104}{3\sqrt[3]{64^2}}, \frac{44}{3\sqrt[3]{64^2}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, 2, ?]$... ??? ... $x + 2z = 0$
 (d) $\Delta x = -0.01$, $\Delta y = -0.1$, $D_0 = [0, 2]$??? -0.05
 (e) $h(x, y) = -7x^2 - 3y^2 + 5xy + 2x$??? $13/59$
 (f) $y = 2x + 3$??? $\left[\frac{-19}{18}, \frac{8}{9}\right]$

2.



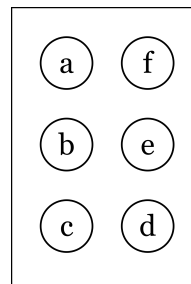
Braille letter

Functions of Two Variables, group *Eta* η -iii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[4]{5x^2y + 9xy^2}$, $\frac{\partial f}{\partial y}$??? $\frac{5x^2-18xy}{4\sqrt[4]{(5x^2y+9xy^2)^3}}$
 (b) $A = [4, 4]$??? $\left(\frac{304}{4\sqrt[4]{896^3}}, \frac{368}{4\sqrt[4]{896^3}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -1, ?]$. ??? . $x + z = 1$
 (d) $\Delta x = -0.2$, $\Delta y = 0.3$, $D_0 = [0, -1]$??? -0.3
 (e) $h(x, y) = x^2 + 3y^2 - xy - 2x$??? -1
 (f) $y = x - 5$??? $\left[\frac{9}{2}, \frac{-1}{2}\right]$

3.



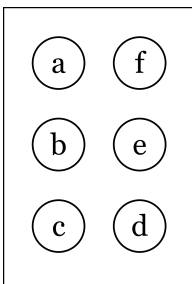
Braille letter

Functions of Two Variables, group *Eta* η -iv

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[4]{6x^2y + xy^2}$, $\frac{\partial f}{\partial x}$.. ??? .. $\frac{12xy+y^2}{4\sqrt[4]{(6x^2y+xy^2)^3}}$
 (b) $A = [3, 1]$??? $\left(\frac{74}{4\sqrt[4]{57^3}}, \frac{60}{4\sqrt[4]{57^3}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^2, 1, ?]$??? $\frac{x}{e^2} + y + z + 2 = 0$
 (d) $\Delta x = 0.3$, $\Delta y = -0.3$, $D_0 = [e^2, 1]$.. ??? .. 1.741
 (e) $h(x, y) = -3x^2 - y^2 - xy + 6x$??? $36/11$
 (f) $y = 2x + 7$??? $\left[\frac{-29}{18}, \frac{34}{9}\right]$

4.



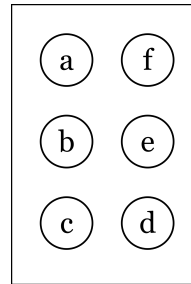
Braille letter

Functions of Two Variables, group *Theta* θ -i

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{4x^2y + xy^2}$, $\frac{\partial f}{\partial x}$... ??? ... $\frac{8xy+y^2}{2\sqrt{4x^2y+xy^2}}$
 (b) $A = [1, 2]$??? $\left(\frac{20}{2\sqrt{12}}, \frac{16}{2\sqrt{12}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^2, 1, ?]$??? $\frac{x}{e^2} + y + z + 2 = 0$
 (d) $\Delta x = 0.1$, $\Delta y = -0.01$, $D_0 = [e^2, 1]$. ??? . 2.004
 (e) $h(x, y) = -6x^2 - 6y^2 - 6xy - x$??? $1/18$
 (f) $y = -3x + 1$??? $\left[\frac{29}{84}, \frac{-1}{28}\right]$

1.



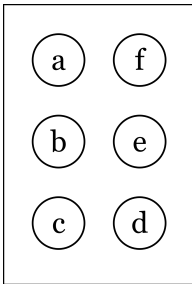
Braille letter

Functions of Two Variables, group *Theta* θ -ii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{2x^2y + 2xy^2}$, $\frac{\partial f}{\partial y}$. ??? . $\frac{2x^2+4xy}{2\sqrt{2x^2y+2xy^2}}$
 (b) $A = [2, 3]$??? $\left(\frac{84}{2\sqrt{60}}, \frac{32}{2\sqrt{60}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -3, ?]$. ??? . $x + 3z = 0$
 (d) $\Delta x = -0.2$, $\Delta y = -0.01$, $D_0 = [0, -3]$??? 0.6
 (e) $h(x, y) = 7x^2 + y^2 + 3xy - x$??? $-1/19$
 (f) $y = -8x - 1$??? $\left[\frac{10}{47}, \frac{1}{47}\right]$

2.



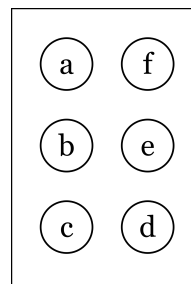
Braille letter

Functions of Two Variables, group *Theta* θ -iii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{2x^2y + 3xy^2}$, $\frac{\partial f}{\partial x}$.. ??? .. $\frac{4xy+3y^2}{2\sqrt{2x^2y+3xy^2}}$
 (b) $A = [3, 3]$??? $\left(\frac{126}{2\sqrt{135}}, \frac{72}{2\sqrt{135}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^4, 1, ?]$??? $\frac{x}{e^4} - y - z + 4 = 0$
 (d) $\Delta x = -0.3$, $\Delta y = 0.2$, $D_0 = [e^4, 1]$.. ??? .. 4.295
 (e) $h(x, y) = -x^2 - 5y^2 + 4xy + x$??? $5/4$
 (f) $y = x - 1$??? $\left[\frac{-13}{4}, \frac{3}{4}\right]$

3.



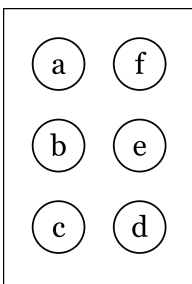
Braille letter

Functions of Two Variables, group *Theta* θ -iv

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[3]{x^2y + xy^2}$, $\frac{\partial f}{\partial x}$... ??? ... $\frac{2xy+y^2}{3\sqrt[3]{(x^2y+xy^2)^2}}$
 (b) $A = [1, 1]$??? $\left(\frac{3}{3\sqrt[3]{2^2}}, \frac{3}{3\sqrt[3]{2^2}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^3, 1, ?]$??? $\frac{x}{e^3} - y - z + 3 = 0$
 (d) $\Delta x = 0.01$, $\Delta y = 0.1$, $D_0 = [e^3, 1]$... ??? ... 3.1
 (e) $h(x, y) = -2x^2 - 3y^2 + xy - 2x$??? $12/23$
 (f) $y = 2x + 5$??? $\left[\frac{21}{8}, \frac{1}{4}\right]$

4.



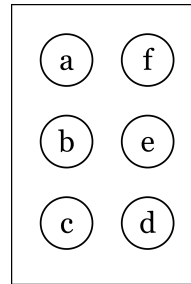
Braille letter

Functions of Two Variables, group *Iota* ι -i

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[3]{5x^2y + 2xy^2}$, $\frac{\partial f}{\partial y}$??? $\frac{5x^2+4xy}{3\sqrt[3]{(5x^2y+2xy^2)^2}}$
 (b) $A = [2, 1]$??? $\left(\frac{22}{3\sqrt[3]{24^2}}, \frac{28}{3\sqrt[3]{24^2}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e, 1, ?]$??? $\frac{x}{e} - y - z + 2 = 0$
 (d) $\Delta x = 0.2$, $\Delta y = 0.01$, $D_0 = [e, 1]$.. ??? .. 0.874
 (e) $h(x, y) = -3x^2 - y^2 - 2xy + 4x$??? 1
 (f) $y = x + 2$??? $\left[-\frac{1}{3}, \frac{5}{3}\right]$

1.



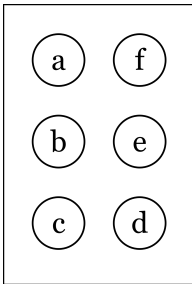
Braille letter

Functions of Two Variables, group *Iota* ι -ii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[4]{6x^2y + 4xy^2}$, $\frac{\partial f}{\partial x}$??? $\frac{12xy+4y^2}{4\sqrt[4]{(6x^2y+4xy^2)^3}}$
 (b) $A = [2, 4]$??? $\left(\frac{160}{4\sqrt[4]{224^3}}, \frac{176}{4\sqrt[4]{224^3}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, 1, ?]$.. ??? .. $x + z = 0$
 (d) $\Delta x = 0.01$, $\Delta y = 0.03$, $D_0 = [0, 1]$. ??? . 0.03
 (e) $h(x, y) = -2x^2 - 7y^2 - xy + 3x$??? $21/19$
 (f) $y = -2x + 2$??? $\left[-\frac{55}{56}, \frac{-1}{28}\right]$

2.



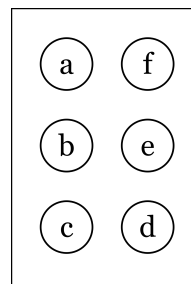
Braille letter

Functions of Two Variables, group *Iota* ι -iii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[3]{2x^2y + 8xy^2}$, $\frac{\partial f}{\partial y}$??? $\frac{2x^2+16xy}{3\sqrt[3]{(2x^2y+8xy^2)^2}}$
 (b) $A = [3, 3]$??? $\left(\frac{108}{3\sqrt[3]{270^2}}, \frac{162}{3\sqrt[3]{270^2}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -1, ?]$. ??? . $x + z = 0$
 (d) $\Delta x = 0.02$, $\Delta y = 0.2$, $D_0 = [0, -1]$??? -0.2
 (e) $h(x, y) = 5x^2 + 7y^2 - 2xy + x$??? $-7/144$
 (f) $y = -2x - 2$??? $\left[-\frac{61}{74}, \frac{-13}{29}\right]$

3.



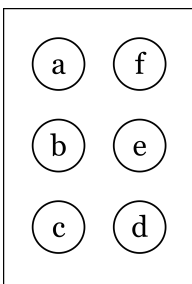
Braille letter

Functions of Two Variables, group *Iota* ι -iv

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[4]{7x^2y + 5xy^2}$, $\frac{\partial f}{\partial x}$. ??? . $\frac{14xy+5y^2}{4\sqrt[4]{(7x^2y+5xy^2)^3}}$
 (b) $A = [2, 4]$??? $\left(\frac{192}{4\sqrt[4]{272^3}}, \frac{108}{4\sqrt[4]{272^3}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^2, 1, ?]$??? $\frac{x}{e^2} - y - z + 2 = 0$
 (d) $\Delta x = -0.1$, $\Delta y = 0.01$, $D_0 = [e^2, 1]$. ??? . 2.086
 (e) $h(x, y) = 4x^2 + y^2 + 3xy + 6x$??? $-36/25$
 (f) $y = -3x - 6$??? $[6, 3]$

4.



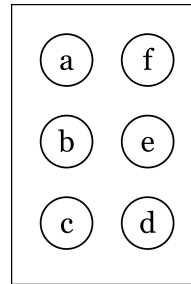
Braille letter

Functions of Two Variables, group *Kappa* κ -i

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{2x^2y + 7xy^2}$, $\frac{\partial f}{\partial x}$.. ??? .. $\frac{4xy+7y^2}{2\sqrt{2x^2y+7xy^2}}$
 (b) $A = [1, 3]$??? $\left(\frac{75}{2\sqrt{69}}, \frac{88}{2\sqrt{69}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^4, 1, ?]$??? $\frac{x}{e^4} + y + z + 4 = 0$
 (d) $\Delta x = -0.1$, $\Delta y = 0.3$, $D_0 = [e^4, 1]$.. ??? .. 4.098
 (e) $h(x, y) = 2x^2 + 6y^2 - 5xy - x$??? $-6/23$
 (f) $y = 2x + 3$??? $\left[\frac{-7}{4}, \frac{-1}{2}\right]$

1.



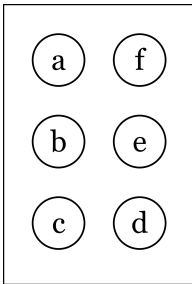
Braille letter

Functions of Two Variables, group *Kappa* κ -ii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[3]{4x^2y + 9xy^2}$, $\frac{\partial f}{\partial y}$??? $\frac{4x^2+18xy}{3\sqrt[3]{(4x^2y+9xy^2)^2}}$
 (b) $A = [1, 1]$??? $\left(\frac{17}{3\sqrt[3]{13^2}}, \frac{44}{3\sqrt[3]{13^2}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, 3, ?]$.. ??? .. $x - 3z = 1$
 (d) $\Delta x = 0.03$, $\Delta y = 0.3$, $D_0 = [0, 3]$.. ??? .. 0.1
 (e) $h(x, y) = 6x^2 + 3y^2 - 4xy + x$??? $-3/56$
 (f) $y = 2x - 4$??? $\left[\frac{-13}{4}, \frac{-9}{10}\right]$

2.



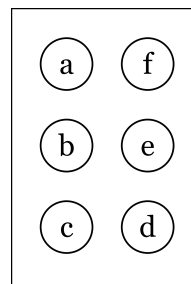
Braille letter

Functions of Two Variables, group *Kappa* κ -iii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{6x^2y + 5xy^2}$, $\frac{\partial f}{\partial x}$.. ??? .. $\frac{12xy-5y^2}{2\sqrt{6x^2y+5xy^2}}$
 (b) $A = [2, 2]$??? $\left(\frac{68}{2\sqrt{88}}, \frac{64}{2\sqrt{88}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [1, 1, ?]$??? $x - y - z + 0 = 0$
 (d) $\Delta x = 0.02$, $\Delta y = 0.02$, $D_0 = [1, 1]$.. ??? .. 0.04
 (e) $h(x, y) = -x^2 - 4y^2 + 2xy + 2x$??? $4/5$
 (f) $y = x - 6$??? $\left[\frac{19}{3}, \frac{1}{3}\right]$

3.



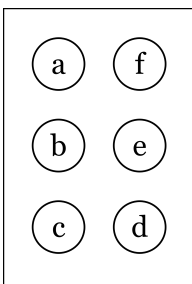
Braille letter

Functions of Two Variables, group *Kappa* κ -iv

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[3]{5x^2y + xy^2}$, $\frac{\partial f}{\partial y}$??? $\frac{5x^2+2xy}{3\sqrt[3]{(5x^2y+xy^2)^2}}$
 (b) $A = [4, 4]$??? $\left(\frac{352}{3\sqrt[3]{384^2}}, \frac{112}{3\sqrt[3]{384^2}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -1, ?]$??? $x + z = 0$
 (d) $\Delta x = 0.2$, $\Delta y = -0.1$, $D_0 = [0, -1]$??? 0.1
 (e) $h(x, y) = -2x^2 - 6y^2 - 6xy + 3x$... ??? ... $9/14$
 (f) $y = -x - 2$??? $\left[\frac{-9}{4}, \frac{1}{28}\right]$

4.



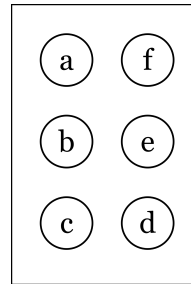
Braille letter

Functions of Two Variables, group *Lambda* λ -i

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[4]{4x^2y + 4xy^2}$, $\frac{\partial f}{\partial y}$??? $\frac{4x^2+4xy}{4\sqrt[4]{(4x^2y+4xy^2)^3}}$
 (b) $A = [4, 4]$??? $\left(\frac{192}{4\sqrt[4]{512^3}}, \frac{192}{4\sqrt[4]{512^3}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -3, ?]$.. ??? .. $x + 3z = 0$
 (d) $\Delta x = -0.02$, $\Delta y = -0.1$, $D_0 = [0, -3]$??? 0.06
 (e) $h(x, y) = -4x^2 - 4y^2 - 7xy - x$??? $4/15$
 (f) $y = x - 1$??? $\left[\frac{7}{15}, \frac{-8}{15}\right]$

1.



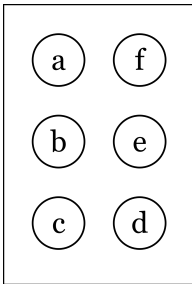
Braille letter

Functions of Two Variables, group *Lambda* λ -ii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[4]{x^2y + 3xy^2}$, $\frac{\partial f}{\partial y}$.. ??? .. $\frac{x^2+6xy}{4\sqrt[4]{(x^2y+3xy^2)^3}}$
 (b) $A = [2, 1]$??? $\left(\frac{7}{4\sqrt[4]{10^3}}, \frac{32}{4\sqrt[4]{10^3}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^4, 1, ?]$??? $\frac{x}{e^4} - y - z + 4 = 0$
 (d) $\Delta x = -0.02$, $\Delta y = -0.03$, $D_0 = [e^4, 1]$??? 4.02
 (e) $h(x, y) = 3x^2 + 4y^2 - 4xy + 3x$??? $-9/8$
 (f) $y = -x + 1$??? $\left[\frac{9}{22}, \frac{13}{6}\right]$

2.



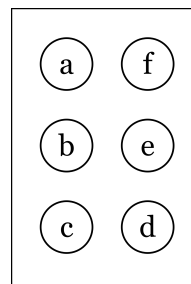
Braille letter

Functions of Two Variables, group *Lambda* λ -iii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{x^2y + xy^2}$, $\frac{\partial f}{\partial y}$??? $\frac{x^2+2xy}{2\sqrt{x^2y+xy^2}}$
 (b) $A = [2, 3]$??? $\left(\frac{21}{2\sqrt{30}}, \frac{16}{2\sqrt{30}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^4, 1, ?]$??? $\frac{x}{e^4} - y - z + 4 = 0$
 (d) $\Delta x = 0.01$, $\Delta y = 0.01$, $D_0 = [e^4, 1]$.. ??? .. 3.99
 (e) $h(x, y) = 3x^2 + 5y^2 - 3xy + x$??? $-5/51$
 (f) $y = x + 2$??? $\left[\frac{-3}{2}, \frac{5}{22}\right]$

3.



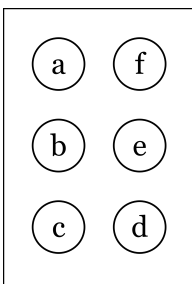
Braille letter

Functions of Two Variables, group *Lambda* λ -iv

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{3x^2y + xy^2}$, $\frac{\partial f}{\partial x}$.. ??? .. $\frac{6xy+y^2}{2\sqrt{3x^2y+xy^2}}$
 (b) $A = [2, 2]$??? $\left(\frac{28}{2\sqrt{32}}, \frac{40}{2\sqrt{32}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -3, ?]$. ??? . $x + 3z = 0$
 (d) $\Delta x = 0.1$, $\Delta y = -0.01$, $D_0 = [0, -3]$??? 0.003
 (e) $h(x, y) = -3x^2 - 3y^2 - xy - x$??? $3/35$
 (f) $y = -2x + 1$??? $\left[\frac{5}{13}, \frac{3}{13}\right]$

4.



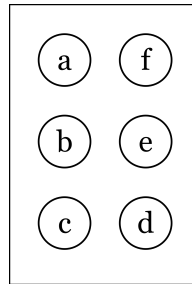
Braille letter

Functions of Two Variables, group $Mu \mu$ -i

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{6x^2y + 4xy^2}$, $\frac{\partial f}{\partial y}$.. ??? .. $\frac{6x^2+4xy}{2\sqrt{6x^2y+4xy^2}}$
 (b) $A = [2, 2]$??? $\left(\frac{64}{2\sqrt{80}}, \frac{56}{2\sqrt{80}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [1, 1, ?]$??? $x - y - z + 0 = 0$
 (d) $\Delta x = -0.1$, $\Delta y = -0.03$, $D_0 = [1, 1]$. ??? . 0.0
 (e) $h(x, y) = -2x^2 - y^2 - 2xy - 2x$??? 1
 (f) $y = 4x - 1$??? $\left[\frac{2}{13}, \frac{-5}{13}\right]$

1.



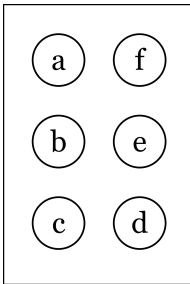
Braille letter

Functions of Two Variables, group $Mu \mu$ -ii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{2x^2y + xy^2}$, $\frac{\partial f}{\partial y}$.. ??? .. $\frac{2x^2+2xy}{2\sqrt{2x^2y+xy^2}}$
 (b) $A = [3, 1]$??? $\left(\frac{26}{2\sqrt{21}}, \frac{24}{2\sqrt{21}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -2, ?]$. ??? . $x + 2z = 0$
 (d) $\Delta x = 0.2$, $\Delta y = -0.01$, $D_0 = [0, -2]$??? -0.1
 (e) $h(x, y) = 7x^2 + y^2 - 4xy + 7x$??? -4
 (f) $y = -x - 2$??? $\left[\frac{-19}{24}, \frac{-29}{8}\right]$

2.



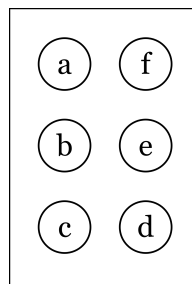
Braille letter

Functions of Two Variables, group $Mu \mu$ -iii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{x^2y + 7xy^2}$, $\frac{\partial f}{\partial y}$.. ??? .. $\frac{x^2+14xy}{2\sqrt{x^2y+7xy^2}}$
 (b) $A = [4, 2]$??? $\left(\frac{44}{2\sqrt{144}}, \frac{128}{2\sqrt{144}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, 1, ?]$.. ??? .. $x - z = 0$
 (d) $\Delta x = 0.2$, $\Delta y = -0.02$, $D_0 = [0, 1]$??? -0.02
 (e) $h(x, y) = -2x^2 - 3y^2 + xy - 3x$??? $27/23$
 (f) $y = 2x + 3$??? $\left[\frac{3}{2}, 0\right]$

3.



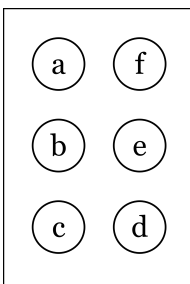
Braille letter

Functions of Two Variables, group $Mu \mu$ -iv

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[3]{3x^2y + 7xy^2}$, $\frac{\partial f}{\partial y}$??? $\frac{3x^2+14xy}{3\sqrt[3]{(3x^2y+7xy^2)^2}}$
 (b) $A = [2, 2]$??? $\left(\frac{52}{3\sqrt[3]{80^2}}, \frac{136}{3\sqrt[3]{80^2}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -3, ?]$. ??? . $x + 3z = 0$
 (d) $\Delta x = 0.01$, $\Delta y = 0.02$, $D_0 = [0, -3]$??? -0.03
 (e) $h(x, y) = -2x^2 - 4y^2 - 4xy - x$??? $1/4$
 (f) $y = -3x - 2$??? $\left[\frac{-41}{52}, \frac{19}{52}\right]$

4.



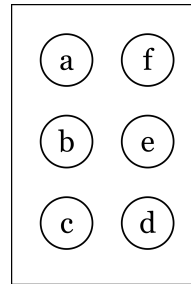
Braille letter

Functions of Two Variables, group $Nu \nu$ -i

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[3]{x^2y + 9xy^2}$, $\frac{\partial f}{\partial x}$... ??? ... $\frac{2xy+9y^2}{3\sqrt[3]{(x^2y+9xy^2)^2}}$
 (b) $A = [4, 4]$??? $\left(\frac{176}{3\sqrt[3]{640^2}}, \frac{608}{3\sqrt[3]{640^2}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^3, 1, ?]$??? $\frac{x}{e^3} + y + z + 3 = 0$
 (d) $\Delta x = 0.1$, $\Delta y = 0.01$, $D_0 = [e^3, 1]$.. ??? .. 3.015
 (e) $h(x, y) = 5x^2 + y^2 - 2xy - 7x$??? $-49/16$
 (f) $y = -2x + 1$??? $[\frac{1}{2}, 0]$

1.



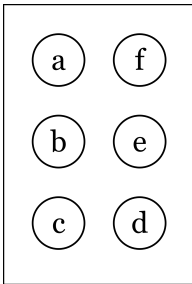
Braille letter

Functions of Two Variables, group $Nu \nu$ -ii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[3]{8x^2y + 3xy^2}$, $\frac{\partial f}{\partial y}$??? $\frac{8x^2-6xy}{3\sqrt[3]{(8x^2y+3xy^2)^2}}$
 (b) $A = [2, 2]$??? $\left(\frac{76}{3\sqrt[3]{88^2}}, \frac{56}{3\sqrt[3]{88^2}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e, 1, ?]$??? $\frac{x}{e} - y - z + 2 = 0$
 (d) $\Delta x = 0.2$, $\Delta y = -0.01$, $D_0 = [e, 1]$. ??? . 1.064
 (e) $h(x, y) = 4x^2 + 6y^2 + xy + x$??? $-1/19$
 (f) $y = x - 6$??? $[\frac{7}{2}, \frac{-5}{2}]$

2.



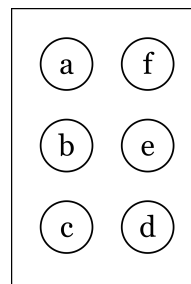
Braille letter

Functions of Two Variables, group $Nu \nu$ -iii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[3]{x^2y + 2xy^2}$, $\frac{\partial f}{\partial x}$??? $\frac{2xy+2y^2}{3\sqrt[3]{(x^2y+2xy^2)^2}}$
 (b) $A = [2, 1]$??? $\left(\frac{6}{3\sqrt[3]{8^2}}, \frac{12}{3\sqrt[3]{8^2}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -1, ?]$. ??? . $x + z = 0$
 (d) $\Delta x = 0.01$, $\Delta y = 0.2$, $D_0 = [0, -1]$??? -0.2
 (e) $h(x, y) = x^2 + y^2 - xy + x$??? $-1/3$
 (f) $y = -2x + 3$??? $[\frac{-5}{7}, 1]$

3.



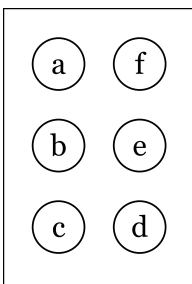
Braille letter

Functions of Two Variables, group $Nu \nu$ -iv

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{8x^2y + 8xy^2}$, $\frac{\partial f}{\partial x}$??? $\frac{16xy-8y^2}{2\sqrt{8x^2y+8xy^2}}$
 (b) $A = [4, 4]$??? $\left(\frac{384}{2\sqrt{1024}}, \frac{384}{2\sqrt{1024}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, 1, ?]$. ??? . $x - z = 0$
 (d) $\Delta x = 0.1$, $\Delta y = 0.03$, $D_0 = [0, 1]$??? 0.03
 (e) $h(x, y) = x^2 + 4y^2 - 2xy - 5x$... ??? ... $-25/3$
 (f) $y = -2x + 2$??? $[\frac{41}{42}, \frac{1}{21}]$

4.



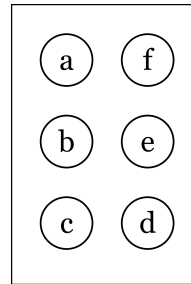
Braille letter

Functions of Two Variables, group Xi ξ -i

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[3]{7x^2y + 3xy^2}$, $\frac{\partial f}{\partial y}$. ??? . $\frac{7x^2+6xy}{3\sqrt[3]{(7x^2y+3xy^2)^2}}$
 (b) $A = [4, 2]$??? $\left(\frac{124}{3\sqrt[3]{27^2}}, \frac{160}{3\sqrt[3]{27^2}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, 3, ?]$... ??? ... $x - 3z = 1$
 (d) $\Delta x = -0.03$, $\Delta y = -0.1$, $D_0 = [0, 3]$??? -0.033
 (e) $h(x, y) = -2x^2 - 4y^2 - 2xy - 4x$??? $16/9$
 (f) $y = 2x - 6$??? $\left[\frac{26}{11}, \frac{-14}{11}\right]$

1.



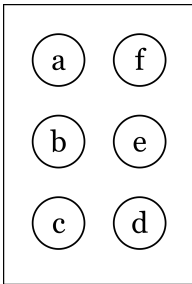
Braille letter

Functions of Two Variables, group Xi ξ -ii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[3]{8x^2y + 3xy^2}$, $\frac{\partial f}{\partial x}$. ??? . $\frac{8xy+3y^2}{3\sqrt[3]{(8x^2y+3xy^2)^2}}$
 (b) $A = [3, 2]$??? $\left(\frac{108}{3\sqrt[3]{180^2}}, \frac{108}{3\sqrt[3]{180^2}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -2, ?]$.. ??? .. $x - 2z = 0$
 (d) $\Delta x = 0.02$, $\Delta y = 0.03$, $D_0 = [0, -2]$??? -0.015
 (e) $h(x, y) = -4x^2 - 2y^2 - 3xy - 5x$??? $50/41$
 (f) $y = x + 5$??? $\left[\frac{-20}{9}, \frac{25}{9}\right]$

2.



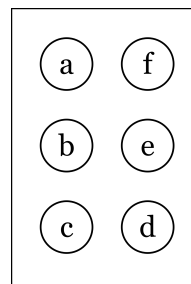
Braille letter

Functions of Two Variables, group Xi ξ -iii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{3x^2y + 2xy^2}$, $\frac{\partial f}{\partial y}$.. ??? .. $\frac{3x^2+4xy}{2\sqrt{3x^2y+2xy^2}}$
 (b) $A = [4, 1]$??? $\left(\frac{26}{2\sqrt{56}}, \frac{64}{2\sqrt{56}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^3, 1, ?]$??? $\frac{x}{e^3} - y - z + 3 = 0$
 (d) $\Delta x = -0.1$, $\Delta y = -0.03$, $D_0 = [e^3, 1]$??? 3.025
 (e) $h(x, y) = -6x^2 - 5y^2 - 7xy - x$??? $5/169$
 (f) $y = x - 2$??? $\left[\frac{-7}{36}, \frac{-13}{12}\right]$

3.



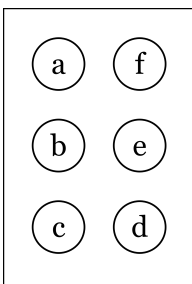
Braille letter

Functions of Two Variables, group Xi ξ -iv

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[3]{2x^2y + xy^2}$, $\frac{\partial f}{\partial y}$.. ??? .. $\frac{2x^2+2xy}{3\sqrt[3]{(2x^2y+xy^2)^2}}$
 (b) $A = [1, 4]$??? $\left(\frac{64}{3\sqrt[3]{24^2}}, \frac{10}{3\sqrt[3]{24^2}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^4, 1, ?]$??? $\frac{x}{e^4} + y + z + 4 = 0$
 (d) $\Delta x = -0.2$, $\Delta y = 0.03$, $D_0 = [e^4, 1]$. ??? . 4.196
 (e) $h(x, y) = -3x^2 - 2y^2 + 3xy + 7x$??? $98/15$
 (f) $y = -5x + 1$??? $\left[\frac{-5}{68}, \frac{-7}{68}\right]$

4.



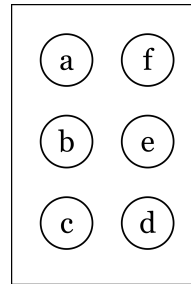
Braille letter

Functions of Two Variables, group *Omicron o* -i

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[4]{3x^2y + 6xy^2}$, $\frac{\partial f}{\partial y}$. ??? . $\frac{3x^2+12xy}{4\sqrt[4]{(3x^2y+6xy^2)^3}}$
 (b) $A = [3, 1]$??? $\left(\frac{24}{4\sqrt[4]{45^3}}, \frac{63}{4\sqrt[4]{45^3}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -3, ?]$.. ??? .. $x + 3z = 1$
 (d) $\Delta x = -0.02$, $\Delta y = 0.2$, $D_0 = [0, -3]$??? -0.067
 (e) $h(x, y) = -3x^2 - 2y^2 - xy - 7x$??? $98/23$
 (f) $y = -7x + 1$??? $\left[\frac{5}{47}, \frac{12}{47}\right]$

1.



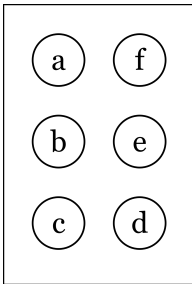
Braille letter

Functions of Two Variables, group *Omicron o* -ii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{6x^2y + 4xy^2}$, $\frac{\partial f}{\partial y}$. ??? . $\frac{6x^2-8xy}{2\sqrt{6x^2y+4xy^2}}$
 (b) $A = [4, 1]$??? $\left(\frac{52}{2\sqrt{112}}, \frac{128}{2\sqrt{112}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, 1, ?]$.. ??? .. $x + z = 0$
 (d) $\Delta x = 0.1$, $\Delta y = -0.03$, $D_0 = [0, 1]$??? -0.03
 (e) $h(x, y) = -2x^2 - 2y^2 + 3xy - 2x$??? $8/25$
 (f) $y = x - 4$??? $[1, -3]$

2.



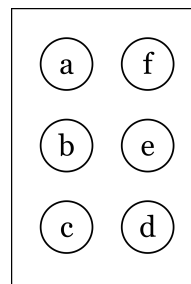
Braille letter

Functions of Two Variables, group *Omicron o* -iii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{4x^2y + 4xy^2}$, $\frac{\partial f}{\partial x}$.. ??? .. $\frac{8xy+4y^2}{2\sqrt{4x^2y+4xy^2}}$
 (b) $A = [2, 4]$??? $\left(\frac{128}{2\sqrt{192}}, \frac{80}{2\sqrt{192}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^4, 1, ?]$??? $\frac{x}{e^4} + y + z + 4 = 0$
 (d) $\Delta x = 0.3$, $\Delta y = 0.1$, $D_0 = [e^4, 1]$.. ??? .. 4.105
 (e) $h(x, y) = 4x^2 + y^2 + xy - x$??? 0
 (f) $y = 4x + 6$??? $\left[\frac{-53}{48}, \frac{19}{12}\right]$

3.



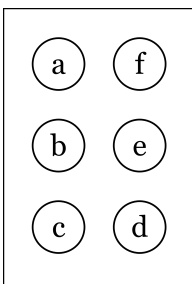
Braille letter

Functions of Two Variables, group *Omicron o* -iv

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{9x^2y + xy^2}$, $\frac{\partial f}{\partial y}$... ??? ... $\frac{9x^2-2xy}{\sqrt{9x^2y+xy^2}}$
 (b) $A = [1, 2]$??? $\left(\frac{40}{2\sqrt{22}}, \frac{13}{2\sqrt{22}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -1, ?]$.. ??? .. $x + z = 0$
 (d) $\Delta x = 0.02$, $\Delta y = -0.02$, $D_0 = [0, -1]$??? 0.02
 (e) $h(x, y) = -3x^2 - 4y^2 + xy + 3x$??? $36/47$
 (f) $y = 2x + 4$??? $\left[\frac{-57}{34}, \frac{11}{17}\right]$

4.



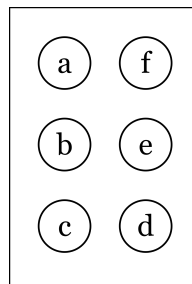
Braille letter

Functions of Two Variables, group $Pi \pi$ -i

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[4]{3x^2y + 6xy^2}$, $\frac{\partial f}{\partial x}$??? $\frac{3xy+6y^2}{4\sqrt[4]{(3x^2y+6xy^2)^3}}$
 (b) $A = [2, 3]$??? $\left(\frac{90}{4\sqrt[4]{144^3}}, \frac{84}{4\sqrt[4]{144^3}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [1, 1, ?]$??? $x - y - z + 0 = 0$
 (d) $\Delta x = -0.02$, $\Delta y = 0.02$, $D_0 = [1, 1]$. ??? . 0.0
 (e) $h(x, y) = 3x^2 + y^2 - xy + 5x$??? $-24/11$
 (f) $y = -3x + 2$??? $\left[\frac{3}{10}, \frac{11}{10}\right]$

1.



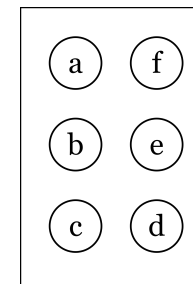
Braille letter

Functions of Two Variables, group $Pi \pi$ -ii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{4x^2y + 6xy^2}$, $\frac{\partial f}{\partial x}$. ??? . $\frac{8xy-6y^2}{2\sqrt{4x^2y+6xy^2}}$
 (b) $A = [2, 4]$??? $\left(\frac{160}{2\sqrt{256}}, \frac{112}{2\sqrt{256}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -2, ?]$. ??? . $x + 2z = 0$
 (d) $\Delta x = 0.03$, $\Delta y = 0.1$, $D_0 = [0, -2]$??? -0.05
 (e) $h(x, y) = 3x^2 + y^2 + xy + 3x$??? $-9/11$
 (f) $y = 2x + 4$??? $\left[\frac{-23}{18}, \frac{13}{9}\right]$

2.



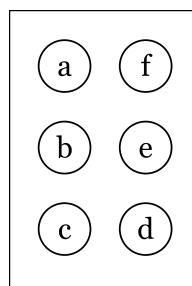
Braille letter

Functions of Two Variables, group $Pi \pi$ -iii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[3]{2x^2y + 3xy^2}$, $\frac{\partial f}{\partial x}$??? $\frac{4xy+3y^2}{3\sqrt[3]{(2x^2y+3xy^2)^2}}$
 (b) $A = [3, 4]$??? $\left(\frac{96}{3\sqrt[3]{216^2}}, \frac{180}{3\sqrt[3]{216^2}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -1, ?]$. ??? . $x - z = 0$
 (d) $\Delta x = -0.2$, $\Delta y = -0.1$, $D_0 = [0, -1]$??? 0.1
 (e) $h(x, y) = -4x^2 - 3y^2 - 6xy - 3x$??? $7/3$
 (f) $y = 3x - 3$??? $\left[\frac{-39}{98}, \frac{-87}{98}\right]$

3.



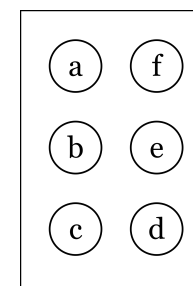
Braille letter

Functions of Two Variables, group $Pi \pi$ -iv

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{7x^2y + 6xy^2}$, $\frac{\partial f}{\partial x}$. ??? . $\frac{14xy+6y^2}{2\sqrt{7x^2y+6xy^2}}$
 (b) $A = [3, 2]$??? $\left(\frac{108}{2\sqrt{198}}, \frac{135}{2\sqrt{198}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -3, ?]$. ??? . $x + 3z = 0$
 (d) $\Delta x = -0.02$, $\Delta y = 0.01$, $D_0 = [0, -3]$??? 0.06
 (e) $h(x, y) = 5x^2 + 3y^2 + 7xy - x$??? $-3/11$
 (f) $y = 4x + 1$??? $\left[\frac{-5}{27}, \frac{21}{25}\right]$

4.



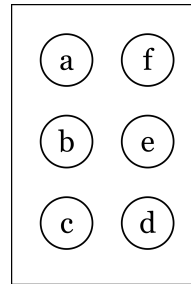
Braille letter

Functions of Two Variables, group *Rho* ρ -i

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{3x^2y + 7xy^2}$, $\frac{\partial f}{\partial x}$... ??? ... $\frac{3xy+7y^2}{2\sqrt{3x^2y+7xy^2}}$
 (b) $A = [1, 4]$??? $\left(\frac{136}{2\sqrt{124}}, \frac{59}{2\sqrt{124}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -1, ?]$.. ??? .. $x + z = 0$
 (d) $\Delta x = -0.1$, $\Delta y = 0.03$, $D_0 = [0, -1]$??? -0.03
 (e) $h(x, y) = 6x^2 + y^2 + xy + 3x$??? $-8/23$
 (f) $y = x + 3$??? $\left[-\frac{3}{4}, \frac{9}{4}\right]$

1.



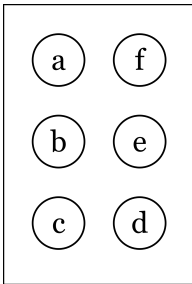
Braille letter

Functions of Two Variables, group *Rho* ρ -ii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[3]{9x^2y + 3xy^2}$, $\frac{\partial f}{\partial x}$??? $\frac{18xy+3y^2}{3\sqrt[3]{(9x^2y+3xy^2)^2}}$
 (b) $A = [1, 1]$??? $\left(\frac{21}{3\sqrt[3]{12^2}}, \frac{15}{3\sqrt[3]{12^2}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [1, 1, ?]$??? $x + y + z + 0 = 0$
 (d) $\Delta x = 0.1$, $\Delta y = 0.02$, $D_0 = [1, 1]$.. ??? .. 0.12
 (e) $h(x, y) = x^2 + 5y^2 + 3xy + 2x$??? $-20/11$
 (f) $y = 2x + 3$??? $\left[\frac{49}{54}, \frac{10}{27}\right]$

2.



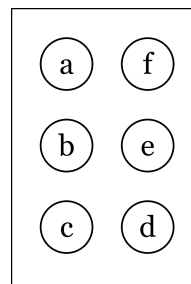
Braille letter

Functions of Two Variables, group *Rho* ρ -iii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{4x^2y + 7xy^2}$, $\frac{\partial f}{\partial x}$... ??? ... $\frac{8xy+7y^2}{2\sqrt{4x^2y+7xy^2}}$
 (b) $A = [1, 4]$??? $\left(\frac{144}{2\sqrt{128}}, \frac{120}{2\sqrt{128}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [1, 1, ?]$??? $x - y - z + 0 = 0$
 (d) $\Delta x = 0.02$, $\Delta y = 0.03$, $D_0 = [1, 1]$.. ??? .. 0.05
 (e) $h(x, y) = -7x^2 - 7y^2 - 5xy + 2x$... ??? ... $28/171$
 (f) $y = 2x - 3$??? $\left[-\frac{67}{90}, -\frac{34}{45}\right]$

3.



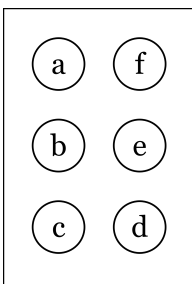
Braille letter

Functions of Two Variables, group *Rho* ρ -iv

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{3x^2y + xy^2}$, $\frac{\partial f}{\partial y}$... ??? ... $\frac{3x^2+2xy}{2\sqrt{3x^2y+xy^2}}$
 (b) $A = [1, 2]$??? $\left(\frac{16}{2\sqrt{10}}, \frac{7}{2\sqrt{10}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -3, ?]$.. ??? .. $x + 3z = 0$
 (d) $\Delta x = -0.02$, $\Delta y = -0.03$, $D_0 = [0, -3]$??? 0.06
 (e) $h(x, y) = -6x^2 - 2y^2 - xy - 7x$??? $99/47$
 (f) $y = -5x + 1$??? $\left[\frac{2}{17}, \frac{7}{17}\right]$

4.



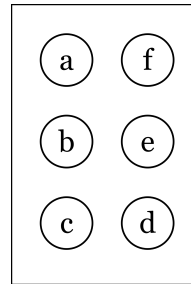
Braille letter

Functions of Two Variables, group *Sigma* σ -i

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{2x^2y + 5xy^2}$, $\frac{\partial f}{\partial y}$??? $\frac{2x^2+10xy}{2\sqrt{2x^2y+5xy^2}}$
 (b) $A = [4, 2]$??? $\left(\frac{52}{2\sqrt{144}}, \frac{112}{2\sqrt{144}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, 1, ?]$.. ??? .. $x - z = 0$
 (d) $\Delta x = -0.2$, $\Delta y = -0.1$, $D_0 = [0, 1]$??? -0.2
 (e) $h(x, y) = 4x^2 + 2y^2 - 3xy - 2x$??? $-8/23$
 (f) $y = x - 7$??? $\left[\frac{-47}{6}, \frac{-11}{2}\right]$

1.



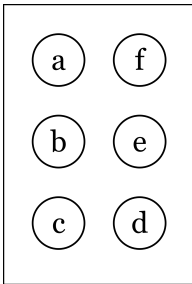
Braille letter

Functions of Two Variables, group *Sigma* σ -ii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[4]{2x^2y + 4xy^2}$, $\frac{\partial f}{\partial y}$??? $\frac{2x^2+8xy}{4\sqrt[4]{(2x^2y+4xy^2)^3}}$
 (b) $A = [3, 1]$??? $\left(\frac{32}{4\sqrt[4]{30^3}}, \frac{42}{4\sqrt[4]{30^3}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -2, ?]$. ??? . $x + 2z = 0$
 (d) $\Delta x = -0.1$, $\Delta y = -0.2$, $D_0 = [0, -2]$??? 0.2
 (e) $h(x, y) = 6x^2 + 5y^2 - 5xy - x$??? $-1/19$
 (f) $y = -x + 3$??? $\left[\frac{-7}{16}, \frac{25}{16}\right]$

2.



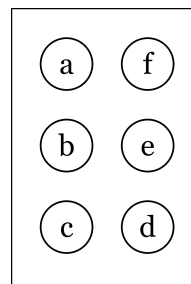
Braille letter

Functions of Two Variables, group *Sigma* σ -iii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[3]{8x^2y + 5xy^2}$, $\frac{\partial f}{\partial y}$. ??? . $\frac{8x^2+10xy}{3\sqrt[3]{(8x^2y+5xy^2)^2}}$
 (b) $A = [2, 3]$??? $\left(\frac{282}{3\sqrt[3]{186^2}}, \frac{92}{3\sqrt[3]{186^2}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^4, 1, ?]$??? $\frac{x}{e^4} - y - z + 4 = 0$
 (d) $\Delta x = 0.03$, $\Delta y = -0.2$, $D_0 = [e^4, 1]$. ??? . 3.971
 (e) $h(x, y) = 4x^2 + 7y^2 - 3xy - 2x$??? $-28/103$
 (f) $y = -3x - 1$??? $\left[\frac{-43}{152}, \frac{-23}{116}\right]$

3.



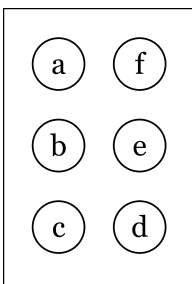
Braille letter

Functions of Two Variables, group *Sigma* σ -iv

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[4]{x^2y + 2xy^2}$, $\frac{\partial f}{\partial x}$.. ??? .. $\frac{2xy+2y^2}{4\sqrt[4]{(x^2y+2xy^2)^3}}$
 (b) $A = [2, 2]$??? $\left(\frac{16}{4\sqrt[4]{24^3}}, \frac{40}{4\sqrt[4]{24^3}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -1, ?]$... ??? ... $x + z = 1$
 (d) $\Delta x = -0.01$, $\Delta y = -0.02$, $D_0 = [0, -1]$??? 0.02
 (e) $h(x, y) = 6x^2 + 3y^2 + 3xy - x$??? $-1/21$
 (f) $y = -3x - 2$??? $\left[\frac{-29}{48}, \frac{-3}{16}\right]$

4.



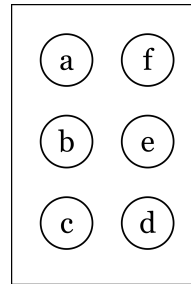
Braille letter

Functions of Two Variables, group *Tau* τ -i

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{3x^2y + 2xy^2}$, $\frac{\partial f}{\partial y}$... ??? ... $\frac{3x^2+4xy}{2\sqrt{3x^2y+2xy^2}}$
 (b) $A = [3, 4]$??? $\left(\frac{104}{2\sqrt{204}}, \frac{150}{2\sqrt{204}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^4, 1, ?]$??? $\frac{x}{e^4} + y + z + 4 = 0$
 (d) $\Delta x = 0.03$, $\Delta y = -0.03$, $D_0 = [e^4, 1]$??? 3.971
 (e) $h(x, y) = -3x^2 - 4y^2 - 4xy + 3x$??? $37/32$
 (f) $y = 4x + 1$??? $\left[\frac{-33}{166}, \frac{17}{83}\right]$

1.



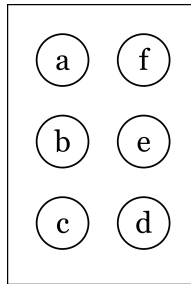
Braille letter

Functions of Two Variables, group *Tau* τ -ii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[4]{2x^2y + 6xy^2}$, $\frac{\partial f}{\partial x}$??? $\frac{4xy+6y^2}{4\sqrt[4]{(2x^2y+6xy^2)^3}}$
 (b) $A = [4, 3]$??? $\left(\frac{102}{4\sqrt[4]{312^3}}, \frac{352}{4\sqrt[4]{312^3}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, 1, ?]$... ??? ... $x + z = 0$
 (d) $\Delta x = -0.02$, $\Delta y = -0.2$, $D_0 = [0, 1]$??? -0.2
 (e) $h(x, y) = x^2 + 7y^2 + 4xy - 2x$??? $-7/3$
 (f) $y = 2x + 2$??? $\left[\frac{-31}{37}, \frac{4}{7}\right]$

2.



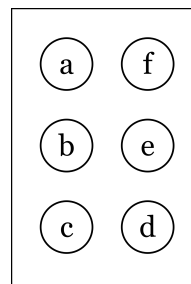
Braille letter

Functions of Two Variables, group *Tau* τ -iii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[3]{3x^2y + 7xy^2}$, $\frac{\partial f}{\partial x}$??? $\frac{6xy+7y^2}{3\sqrt[3]{(3x^2y+7xy^2)^2}}$
 (b) $A = [4, 4]$??? $\left(\frac{208}{3\sqrt[3]{640^2}}, \frac{272}{3\sqrt[3]{640^2}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -1, ?]$.. ??? .. $x + z = 0$
 (d) $\Delta x = -0.1$, $\Delta y = -0.01$, $D_0 = [0, -1]$??? 0.01
 (e) $h(x, y) = 5x^2 + 3y^2 + 6xy + 3x$??? $-13/12$
 (f) $y = -x - 6$??? $\left[\frac{-3}{4}, \frac{-3}{4}\right]$

3.



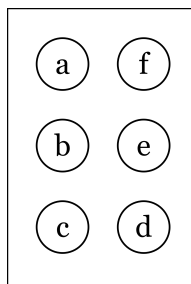
Braille letter

Functions of Two Variables, group *Tau* τ -iv

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[3]{2x^2y + 2xy^2}$, $\frac{\partial f}{\partial x}$??? $\frac{4xy+2y^2}{3\sqrt[3]{(2x^2y+2xy^2)^2}}$
 (b) $A = [4, 4]$??? $\left(\frac{96}{3\sqrt[3]{256^2}}, \frac{96}{3\sqrt[3]{256^2}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -2, ?]$. ??? . $x + 2z = 0$
 (d) $\Delta x = 0.2$, $\Delta y = -0.01$, $D_0 = [0, -2]$??? 0.005
 (e) $h(x, y) = -5x^2 - 7y^2 - 6xy - 2x$??? $7/44$
 (f) $y = 2x - 1$??? $\left[\frac{16}{45}, \frac{-13}{21}\right]$

4.



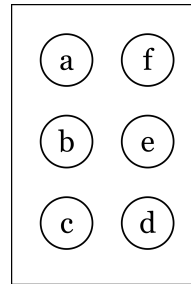
Braille letter

Functions of Two Variables, group *Upsilon v* -i

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{x^2 y + 6xy^2}$, $\frac{\partial f}{\partial y}$.. ??? .. $\frac{x^2 + 12xy}{2\sqrt{x^2 y + 6xy^2}}$
 (b) $A = [2, 1]$??? $\left(\frac{10}{2\sqrt[4]{16}}, \frac{28}{2\sqrt[4]{16}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -2, ?]$. ??? . $x - 2z = 0$
 (d) $\Delta x = 0.01$, $\Delta y = 0.01$, $D_0 = [0, -2]$??? -0.02
 (e) $h(x, y) = 2x^2 + 2y^2 - 3xy - 2x$??? -1
 (f) $y = x - 2$??? $[2, 0]$

1.



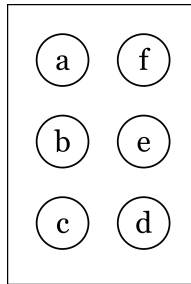
Braille letter

Functions of Two Variables, group *Upsilon v* -ii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[4]{6x^2 y + 2xy^2}$, $\frac{\partial f}{\partial y}$. ??? . $\frac{6x^2 + 4xy}{4\sqrt[4]{(6x^2 y + 2xy^2)^3}}$
 (b) $A = [2, 4]$??? $\left(\frac{128}{4\sqrt[4]{160^3}}, \frac{112}{4\sqrt[4]{160^3}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^4, 1, ?]$??? $\frac{x}{e^4} - y - z + 4 = 0$
 (d) $\Delta x = 0.2$, $\Delta y = -0.01$, $D_0 = [e^4, 1]$. ??? . 3.994
 (e) $h(x, y) = 2x^2 + 5y^2 + xy - 2x$??? $-20/39$
 (f) $y = -2x - 1$??? $\left[\frac{-17}{40}, \frac{-1}{8}\right]$

2.



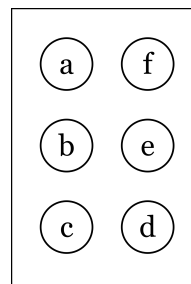
Braille letter

Functions of Two Variables, group *Upsilon v* -iii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[4]{5x^2 y + 2xy^2}$, $\frac{\partial f}{\partial y}$. ??? . $\frac{5x^2 + 4xy}{4\sqrt[4]{(5x^2 y + 2xy^2)^3}}$
 (b) $A = [4, 1]$??? $\left(\frac{42}{4\sqrt[4]{88^3}}, \frac{192}{4\sqrt[4]{88^3}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^2, 1, ?]$??? $\frac{x}{e^2} - y - z + 2 = 0$
 (d) $\Delta x = -0.01$, $\Delta y = 0.02$, $D_0 = [e^2, 1]$??? 2.009
 (e) $h(x, y) = 3x^2 + 3y^2 + xy + 2x$??? $-12/35$
 (f) $y = -2x - 1$??? $\left[\frac{-1}{2}, 0\right]$

3.



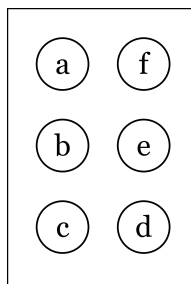
Braille letter

Functions of Two Variables, group *Upsilon v* -iv

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{7x^2 y + 5xy^2}$, $\frac{\partial f}{\partial y}$.. ??? .. $\frac{7x^2 + 10xy}{2\sqrt{7x^2 y + 5xy^2}}$
 (b) $A = [2, 1]$??? $\left(\frac{33}{2\sqrt[4]{38}}, \frac{96}{2\sqrt[4]{38}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [1, 1, ?]$??? $x - y - z + 1 = 0$
 (d) $\Delta x = 0.3$, $\Delta y = -0.03$, $D_0 = [1, 1]$.. ??? .. 0.0
 (e) $h(x, y) = 7x^2 + 6y^2 - 2xy + 4x$??? $-24/41$
 (f) $y = x + 7$??? $\left[\frac{-37}{11}, \frac{40}{11}\right]$

4.



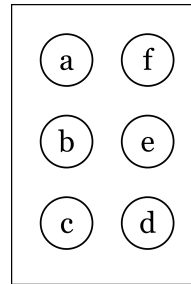
Braille letter

Functions of Two Variables, group *Phi* ϕ -i

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[4]{8x^2y + 3xy^2}$, $\frac{\partial f}{\partial y}$. ??? . $\frac{8x^2+6xy}{3 \sqrt[4]{(8x^2y+3xy^2)^3}}$
 (b) $A = [1, 2]$??? $\left(\frac{44}{4 \sqrt[4]{28^3}}, \frac{20}{4 \sqrt[4]{28^3}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^2, 1, ?]$??? $\frac{x}{e^2} + y + z + 2 = 0$
 (d) $\Delta x = -0.2$, $\Delta y = 0.03$, $D_0 = [e^2, 1]$. ??? . 2.003
 (e) $h(x, y) = -7x^2 - 3y^2 - 6xy - 5x$??? $5/8$
 (f) $y = -x - 8$??? $\left[\frac{91}{8}, \frac{-59}{8}\right]$

1.



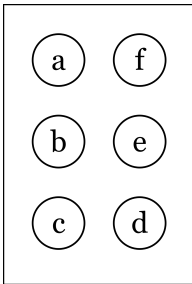
Braille letter

Functions of Two Variables, group *Phi* ϕ -ii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{4x^2y + 9xy^2}$, $\frac{\partial f}{\partial x}$??? $\frac{8xy+9y^2}{2 \sqrt{4x^2y+9xy^2}}$
 (b) $A = [2, 1]$??? $\left(\frac{25}{2 \sqrt[4]{34}}, \frac{104}{2 \sqrt[4]{34}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -1, ?]$. ??? . $x + z = 0$
 (d) $\Delta x = -0.1$, $\Delta y = -0.2$, $D_0 = [0, -1]$??? 0.2
 (e) $h(x, y) = 4x^2 + y^2 - 2xy + 8x$??? $-16/3$
 (f) $y = 2x - 2$??? $\left[\frac{-5}{2}, -3\right]$

2.



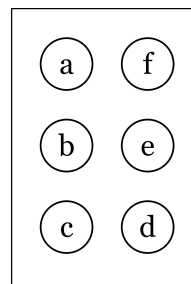
Braille letter

Functions of Two Variables, group *Phi* ϕ -iii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[3]{2x^2y + 2xy^2}$, $\frac{\partial f}{\partial x}$??? $\frac{4xy+2y^2}{3 \sqrt[3]{(2x^2y+2xy^2)^2}}$
 (b) $A = [2, 4]$??? $\left(\frac{64}{3 \sqrt[3]{96^2}}, \frac{80}{3 \sqrt[3]{96^2}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e, 1, ?]$??? $\frac{x}{e} - y - z + 2 = 0$
 (d) $\Delta x = 0.2$, $\Delta y = 0.02$, $D_0 = [e, 1]$.. ??? .. 0.874
 (e) $h(x, y) = 2x^2 + 6y^2 + 3xy + 2x$??? $-8/13$
 (f) $y = -4x + 1$??? $\left[\frac{1}{4}, 0\right]$

3.



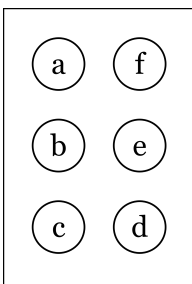
Braille letter

Functions of Two Variables, group *Phi* ϕ -iv

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[4]{3x^2y + 7xy^2}$, $\frac{\partial f}{\partial x}$??? $\frac{6xy+7y^2}{4 \sqrt[4]{(3x^2y+7xy^2)^3}}$
 (b) $A = [2, 4]$??? $\left(\frac{160}{4 \sqrt[4]{272^3}}, \frac{248}{4 \sqrt[4]{272^3}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -2, ?]$.. ??? .. $x + 2z = 0$
 (d) $\Delta x = 0.02$, $\Delta y = -0.3$, $D_0 = [0, -2]$??? -0.01
 (e) $h(x, y) = -6x^2 - 3y^2 + 4xy - 5x$??? $75/56$
 (f) $y = x + 2$??? $\left[\frac{-9}{10}, \frac{11}{10}\right]$

4.



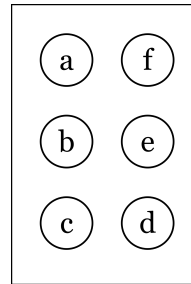
Braille letter

Functions of Two Variables, group *Chi* χ -i

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{x^2 y + 3xy^2}$, $\frac{\partial f}{\partial y}$??? $\frac{x^2 + 6xy}{2\sqrt{x^2 y + 3xy^2}}$
 (b) $A = [3, 3]$??? $\left(\frac{45}{2\sqrt{108}}, \frac{63}{2\sqrt{108}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, 2, ?]$??? $x - 2z = 0$
 (d) $\Delta x = 0.2$, $\Delta y = 0.1$, $D_0 = [0, 2]$??? 0.1
 (e) $h(x, y) = 6x^2 + 2y^2 - 4xy - x$.. ??? .. $-1/16$
 (f) $y = x - 6$??? $\left[\frac{-47}{8}, \frac{-47}{8}\right]$

1.



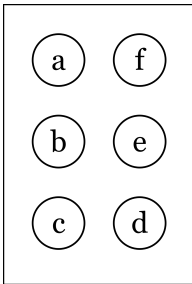
Braille letter

Functions of Two Variables, group *Chi* χ -ii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[3]{4x^2 y + 8xy^2}$, $\frac{\partial f}{\partial y}$. ??? . $\frac{4x^2 + 16xy}{3\sqrt[3]{(4x^2 y + 8xy^2)^2}}$
 (b) $A = [4, 1]$??? $\left(\frac{80}{3\sqrt[3]{96^2}}, \frac{128}{3\sqrt[3]{96^2}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^2, 1, ?]$??? $\frac{x}{e^2} - y - z + 2 = 0$
 (d) $\Delta x = 0.03$, $\Delta y = 0.03$, $D_0 = [e^2, 1]$. ??? . 2.034
 (e) $h(x, y) = x^2 + 5y^2 + 4xy + 2x$??? -5
 (f) $y = x + 5$??? $\left[\frac{7}{5}, \frac{7}{5}\right]$

2.



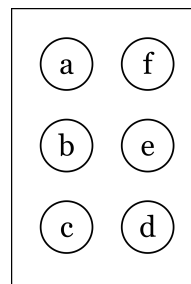
Braille letter

Functions of Two Variables, group *Chi* χ -iii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[4]{7x^2 y + 7xy^2}$, $\frac{\partial f}{\partial x}$. ??? . $\frac{14xy + 7y^2}{4\sqrt[4]{(7x^2 y + 7xy^2)^3}}$
 (b) $A = [2, 3]$??? $\left(\frac{147}{4\sqrt[4]{210^3}}, \frac{112}{4\sqrt[4]{210^3}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^5, 1, ?]$??? $\frac{x}{e^5} - y - z + 5 = 0$
 (d) $\Delta x = 0.03$, $\Delta y = 0.3$, $D_0 = [e^5, 1]$... ??? ... 5.3
 (e) $h(x, y) = -4x^2 - 5y^2 - 3xy - x$??? $5/71$
 (f) $y = x - 1$??? $\left[\frac{1}{2}, -1\right]$

3.



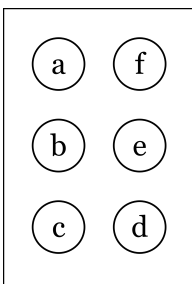
Braille letter

Functions of Two Variables, group *Chi* χ -iv

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{x^2 y + 3xy^2}$, $\frac{\partial f}{\partial y}$.. ??? .. $\frac{x^2 + 6xy}{2\sqrt{x^2 y + 3xy^2}}$
 (b) $A = [1, 3]$??? $\left(\frac{33}{2\sqrt{30}}, \frac{38}{2\sqrt{30}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -1, ?]$. ??? . $x + z = 0$
 (d) $\Delta x = 0.1$, $\Delta y = 0.03$, $D_0 = [0, -1]$??? -0.03
 (e) $h(x, y) = -x^2 - 5y^2 + 3xy + 4x$??? $80/29$
 (f) $y = x + 1$??? $\left[\frac{17}{6}, \frac{1}{2}\right]$

4.



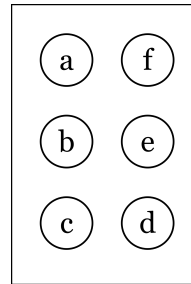
Braille letter

Functions of Two Variables, group *Psi* ψ -i

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{x^2 y + 3xy^2}$, $\frac{\partial f}{\partial x}$. ??? . $\frac{2xy+3y^2}{2\sqrt{x^2 y+3xy^2}}$
 (b) $A = [2, 2]$??? $\left(\frac{20}{2\sqrt{32}}, \frac{28}{2\sqrt{32}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -2, ?]$??? $x - 2z = 0$
 (d) $\Delta x = 0.3$, $\Delta y = 0.3$, $D_0 = [0, -2]$??? -0.3
 (e) $h(x, y) = 3x^2 + 4y^2 + 4xy - 5x$.. ??? .. $-99/32$
 (f) $y = -x - 9$??? $\left[\frac{-31}{6}, \frac{-23}{6}\right]$

1.



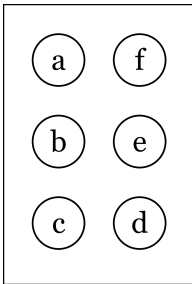
Braille letter

Functions of Two Variables, group *Psi* ψ -ii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[3]{5x^2 y + 4xy^2}$, $\frac{\partial f}{\partial y}$.. ??? .. $\frac{10x^2+8xy}{\sqrt[3]{(5x^2 y+4xy^2)^2}}$
 (b) $A = [1, 2]$??? $\left(\frac{36}{3\sqrt[3]{26^2}}, \frac{21}{3\sqrt[3]{26^2}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^4, 1, ?]$??? $\frac{x}{e^5} - y - z + 5 = 0$
 (d) $\Delta x = 0.03$, $\Delta y = -0.01$, $D_0 = [e^4, 1]$??? 3.991
 (e) $h(x, y) = 6x^2 + 4y^2 + 5xy + 3x$??? $-36/121$
 (f) $y = -x - 5$??? $\left[\frac{-9}{5}, \frac{-16}{5}\right]$

2.



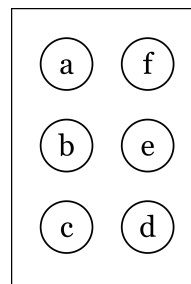
Braille letter

Functions of Two Variables, group *Psi* ψ -iii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[3]{x^2 y + 5xy^2}$, $\frac{\partial f}{\partial y}$. ??? . $\frac{x^2-5xy}{3\sqrt[3]{(x^2 y+5xy^2)^2}}$
 (b) $A = [2, 4]$??? $\left(\frac{96}{3\sqrt[3]{176^2}}, \frac{84}{3\sqrt[3]{176^2}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e, 1, ?]$??? $\frac{x}{e} - y - z + 1 = 0$
 (d) $\Delta x = 0.2$, $\Delta y = -0.2$, $D_0 = [e, 1]$. ??? . 0.874
 (e) $h(x, y) = -6x^2 - 3y^2 + 2xy - 4x$??? $12/19$
 (f) $y = -3x - 3$??? $\left[\frac{-32}{39}, \frac{-7}{13}\right]$

3.



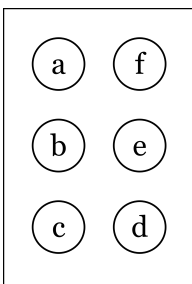
Braille letter

Functions of Two Variables, group *Psi* ψ -iv

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{2x^2 y + xy^2}$, $\frac{\partial f}{\partial y}$. ??? . $\frac{2x^2+2xy}{2\sqrt{2x^2 y+xy^2}}$
 (b) $A = [4, 1]$??? $\left(\frac{17}{2\sqrt{36}}, \frac{40}{2\sqrt{36}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -3, ?]$??? $x + 3z = 1$
 (d) $\Delta x = 0.01$, $\Delta y = 0.3$, $D_0 = [0, -3]$??? -0.1
 (e) $h(x, y) = 5x^2 + y^2 - 2xy - 9x$... ??? ... $-81/16$
 (f) $y = -5x + 1$??? $\left[\frac{1}{80}, \frac{-5}{16}\right]$

4.



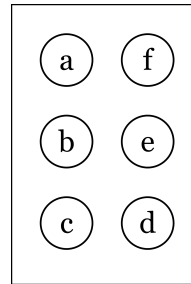
Braille letter

Functions of Two Variables, group *Omega* ω -i

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{8x^2y + 3xy^2}$, $\frac{\partial f}{\partial x}$.. ??? .. $\frac{16xy+3y^2}{2\sqrt{8x^2y+3xy^2}}$
 (b) $A = [1, 4]$??? $\left(\frac{112}{2\sqrt{80}}, \frac{32}{2\sqrt{80}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^2, 1, ?]$??? $\frac{x}{e^2} - y - z + 2 = 0$
 (d) $\Delta x = 0.1$, $\Delta y = -0.3$, $D_0 = [e^2, 1]$.. ??? .. 1.714
 (e) $h(x, y) = -5x^2 - 2y^2 - xy - x$??? $1/13$
 (f) $y = -5x + 1$??? $\left[-\frac{11}{50}, \frac{1}{10}\right]$

1.



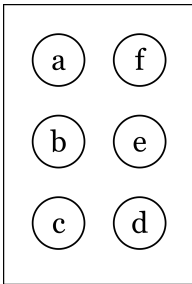
Braille letter

Functions of Two Variables, group *Omega* ω -ii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[4]{5x^2y + xy^2}$, $\frac{\partial f}{\partial y}$.. ??? .. $\frac{5x^2+2xy}{4\sqrt[4]{(5x^2y+xy^2)^3}}$
 (b) $A = [4, 1]$??? $\left(\frac{41}{4\sqrt[4]{84^3}}, \frac{176}{4\sqrt[4]{84^3}}\right)$
 (c) $g(x, y) = \sin \frac{x}{y}$, $D = [0, -2, ?]$.. ??? .. $x - 2z = 0$
 (d) $\Delta x = 0.01$, $\Delta y = -0.03$, $D_0 = [0, -2]$??? 0.015
 (e) $h(x, y) = -x^2 - 3y^2 + xy - x$??? $3/11$
 (f) $y = -2x + 5$??? $\left[-\frac{28}{15}, \frac{11}{15}\right]$

2.



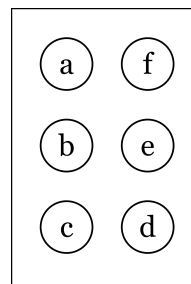
Braille letter

Functions of Two Variables, group *Omega* ω -iii

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt[4]{8x^2y + xy^2}$, $\frac{\partial f}{\partial y}$.. ??? .. $\frac{8x^2+2xy}{4\sqrt[4]{(8x^2y+xy^2)^3}}$
 (b) $A = [4, 4]$??? $\left(\frac{544}{4\sqrt[4]{576^3}}, \frac{160}{4\sqrt[4]{576^3}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^5, 1, ?]$??? $\frac{x}{e^5} + y + z + 5 = 0$
 (d) $\Delta x = 0.2$, $\Delta y = 0.02$, $D_0 = [e^5, 1]$.. ??? .. 4.801
 (e) $h(x, y) = -5x^2 - 2y^2 - 3xy - 7x$??? 2
 (f) $y = x - 7$??? $\left[-\frac{7}{10}, -\frac{49}{10}\right]$

3.



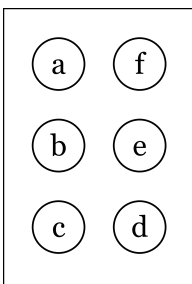
Braille letter

Functions of Two Variables, group *Omega* ω -iv

- (a) Calculate the partial derivative of the given function $f(x, y)$.
 (b) Determine the gradient of function $f(x, y)$ at the given point A .
 (c) Write the equation of the tangent plane to the graph of the function $g(x, y)$ at the point of tangency D .
 (d) Using linear approximation, estimate the value of the function $g(x, y)$ at the point shifted by the vector $(\Delta x, \Delta y)$ from point D_0 .
 (e) Calculate $h(x, y)$ at its local extremum.
 (f) Find the point of the constrained local extremum of the function $h(x, y)$ on the given line $y(x)$.

- (a) $f(x, y) = \sqrt{x^2y + xy^2}$, $\frac{\partial f}{\partial x}$??? $\frac{2xy+y^2}{2\sqrt{x^2y+xy^2}}$
 (b) $A = [1, 3]$??? $\left(\frac{15}{2\sqrt{12}}, \frac{14}{2\sqrt{12}}\right)$
 (c) $g(x, y) = \ln \frac{x}{y}$, $D = [e^2, 1, ?]$??? $\frac{x}{e^2} + y + z + 2 = 0$
 (d) $\Delta x = 0.1$, $\Delta y = 0.3$, $D_0 = [e^2, 1]$.. ??? .. 2.314
 (e) $h(x, y) = 3x^2 + 4y^2 - 5xy + 3x$??? $-36/23$
 (f) $y = 2x + 2$??? $\left[-\frac{25}{18}, -\frac{7}{9}\right]$

4.



Braille letter

Functions of Two Variables (solutions)

α	$i : \mathbf{B}$	(a) $\frac{4x^2 + 2xy}{2\sqrt{4x^2y + xy^2}} \checkmark$	(b) $\left(\frac{80}{2\sqrt{96}}, \frac{32}{2\sqrt{96}}\right) \checkmark$	(c) $\frac{x}{e^4} - y - z + 4 = 0 \mathbf{x}$	(d) $3.7 \mathbf{x}$	(e) $-18/23 \mathbf{x}$	(f) $\left[\frac{37}{12}, \frac{-59}{12}\right] \mathbf{x}$
	$ii : \mathbf{A}$	(a) $\frac{x^2 + 4xy}{4\sqrt{(x^2y + 2xy^2)^3}} \checkmark$	(b) $\left(\frac{24}{4\sqrt[4]{64^3}}, \frac{48}{4\sqrt[4]{64^3}}\right) \mathbf{x}$	(c) $\frac{x}{e^2} - y - z + 2 = 0 \mathbf{x}$	(d) $2.034 \mathbf{x}$	(e) $-3/2 \mathbf{x}$	(f) $\left[\frac{11}{26}, \frac{-7}{26}\right] \mathbf{x}$
	$iii : \mathbf{L}$	(a) $\frac{4xy + y^2}{2\sqrt{2x^2y + xy^2}} \checkmark$	(b) $\left(\frac{64}{2\sqrt{120}}, \frac{42}{2\sqrt{120}}\right) \checkmark$	(c) $x + 3z = 0 \checkmark$	(d) $0.007 \mathbf{x}$	(e) $-16/19 \mathbf{x}$	(f) $\left[\frac{-21}{95}, \frac{11}{95}\right] \mathbf{x}$
	$iv : \mathbf{L}$	(a) $\frac{6x^2 + 8xy}{4\sqrt{(6x^2y + 4xy^2)^3}} \checkmark$	(b) $\left(\frac{108}{4\sqrt[4]{144^3}}, \frac{72}{4\sqrt[4]{144^3}}\right) \checkmark$	(c) $\frac{x}{e^4} - y - z + 4 = 0 \checkmark$	(d) $3.904 \mathbf{x}$	(e) $6/71 \mathbf{x}$	(f) $\left[\frac{16}{43}, \frac{-5}{43}\right] \mathbf{x}$
β	$i : \mathbf{W}$	(a) $\frac{3x^2 + 10xy}{3\sqrt[3]{(3x^2y + 5xy^2)^2}} \checkmark$	(b) $\left(\frac{117}{3\sqrt[3]{324^2}}, \frac{168}{3\sqrt[3]{324^2}}\right) \checkmark$	(c) $x + 3z = 0 \checkmark$	(d) $0.067 \checkmark$	(e) $-12/47 \checkmark$	(f) $\left[\frac{-19}{6}, \frac{29}{6}\right] \mathbf{x}$
	$ii : \mathbf{I}$	(a) $\frac{18xy + 5y^2}{2\sqrt{9x^2y + 5xy^2}} \mathbf{x}$	(b) $\left(\frac{56}{2\sqrt{38}}, \frac{29}{2\sqrt{38}}\right) \checkmark$	(c) $\frac{x}{e^4} - y - z + 4 = 0 \mathbf{x}$	(d) $4.195 \mathbf{x}$	(e) $28/27 \mathbf{x}$	(f) $\left[\frac{30}{31}, \frac{-2}{31}\right] \checkmark$
	$iii : \mathbf{N}$	(a) $\frac{x^2 + 4xy}{2\sqrt{x^2y + 2xy^2}} \checkmark$	(b) $\left(\frac{8}{2\sqrt{15}}, \frac{21}{2\sqrt{15}}\right) \mathbf{x}$	(c) $x - y - z + 0 = 0 \checkmark$	(d) $-0.28 \mathbf{x}$	(e) $-50/23 \checkmark$	(f) $\left[\frac{23}{39}, \frac{-25}{39}\right] \checkmark$
	$iv : \mathbf{G}$	(a) $\frac{x^2 + 6xy}{3\sqrt[3]{(x^2y + 3xy^2)^2}} \checkmark$	(b) $\left(\frac{20}{3\sqrt[3]{32^2}}, \frac{28}{3\sqrt[3]{32^2}}\right) \checkmark$	(c) $\frac{x}{e^3} - y - z + 3 = 0 \mathbf{x}$	(d) $2.97 \mathbf{x}$	(e) $-32/7 \checkmark$	(f) $[0, 4] \checkmark$
γ	$i : \mathbf{H}$	(a) $\frac{6x^2 + 18xy}{2\sqrt{6x^2y + 9xy^2}} \checkmark$	(b) $\left(\frac{153}{2\sqrt{234}}, \frac{132}{2\sqrt{234}}\right) \checkmark$	(c) $x + z = 0 \mathbf{x}$	(d) $-0.01 \mathbf{x}$	(e) $15/13 \checkmark$	(f) $\left[\frac{-29}{16}, \frac{-3}{8}\right] \mathbf{x}$
	$ii : \mathbf{O}$	(a) $\frac{4xy + 3y^2}{2\sqrt{2x^2y + 3xy^2}} \checkmark$	(b) $\left(\frac{20}{2\sqrt{16}}, \frac{14}{2\sqrt{16}}\right) \mathbf{x}$	(c) $x - z = 0 \checkmark$	(d) $-0.03 \mathbf{x}$	(e) $-5/4 \checkmark$	(f) $\left[\frac{-95}{58}, \frac{21}{29}\right] \mathbf{x}$
	$iii : \mathbf{M}$	(a) $\frac{4xy + 7y^2}{3\sqrt[3]{(2x^2y + 7xy^2)^2}} \checkmark$	(b) $\left(\frac{111}{3\sqrt[3]{348^2}}, \frac{200}{3\sqrt[3]{348^2}}\right) \mathbf{x}$	(c) $x + z = 0 \checkmark$	(d) $0.1 \mathbf{x}$	(e) $-1 \mathbf{x}$	(f) $\left[\frac{-2}{7}, \frac{-3}{7}\right] \checkmark$
	$iv : \mathbf{E}$	(a) $\frac{8xy + 3y^2}{2\sqrt{4x^2y + 3xy^2}} \checkmark$	(b) $\left(\frac{76}{2\sqrt{176}}, \frac{112}{2\sqrt{176}}\right) \mathbf{x}$	(c) $x - 2z = 0 \mathbf{x}$	(d) $0.015 \mathbf{x}$	(e) $-45/76 \checkmark$	(f) $\left[\frac{3}{2}, \frac{3}{2}\right] \mathbf{x}$
δ	$i : \mathbf{L}$	(a) $\frac{6xy + 9y^2}{4\sqrt[4]{(3x^2y + 9xy^2)^3}} \checkmark$	(b) $\left(\frac{33}{4\sqrt[4]{84^3}}, \frac{120}{4\sqrt[4]{84^3}}\right) \checkmark$	(c) $\frac{x}{e^4} - y - z + 4 = 0 \checkmark$	(d) $4.3 \mathbf{x}$	(e) $8/39 \mathbf{x}$	(f) $\left[\frac{-21}{100}, \frac{-1}{20}\right] \mathbf{x}$
	$ii : \mathbf{O}$	(a) $\frac{2x^2 + 18xy}{3\sqrt[3]{(2x^2y + 9xy^2)^2}} \checkmark$	(b) $\left(\frac{176}{3\sqrt[3]{320^2}}, \frac{152}{3\sqrt[3]{320^2}}\right) \mathbf{x}$	(c) $x - 2z = 0 \checkmark$	(d) $0.15 \mathbf{x}$	(e) $27/83 \checkmark$	(f) $\left[\frac{-11}{9}, \frac{34}{9}\right] \mathbf{x}$
	$iii : \mathbf{V}$	(a) $\frac{2x^2 + 2xy}{3\sqrt[3]{(2x^2y + xy^2)^2}} \checkmark$	(b) $\left(\frac{21}{3\sqrt[3]{15^2}}, \frac{8}{3\sqrt[3]{15^2}}\right) \checkmark$	(c) $x - y - z + 0 = 0 \checkmark$	(d) $0.0 \checkmark$	(e) $9/25 \mathbf{x}$	(f) $\left[\frac{-69}{26}, \frac{35}{13}\right] \mathbf{x}$
	$iv : \mathbf{E}$	(a) $\frac{5x^2 + 14xy}{4\sqrt[4]{(5x^2y + 7xy^2)^3}} \checkmark$	(b) $\left(\frac{27}{4\sqrt[4]{34^3}}, \frac{48}{4\sqrt[4]{34^3}}\right) \mathbf{x}$	(c) $x - 2z = 0 \mathbf{x}$	(d) $-0.15 \mathbf{x}$	(e) $-8/15 \checkmark$	(f) $\left[\frac{-35}{34}, \frac{-3}{34}\right] \mathbf{x}$
ϵ	$i : \mathbf{W}$	(a) $\frac{12xy + 2y^2}{3\sqrt[3]{(6x^2y + 2xy^2)^2}} \checkmark$	(b) $\left(\frac{224}{3\sqrt[3]{512^2}}, \frac{160}{3\sqrt[3]{512^2}}\right) \checkmark$	(c) $x - 3z = 0 \checkmark$	(d) $0.1 \checkmark$	(e) $1/19 \checkmark$	(f) $\left[\frac{19}{10}, \frac{11}{5}\right] \mathbf{x}$
	$ii : \mathbf{I}$	(a) $\frac{2xy + 3y^2}{2\sqrt{x^2y + 3xy^2}} \mathbf{x}$	(b) $\left(\frac{45}{2\sqrt{108}}, \frac{63}{2\sqrt{108}}\right) \checkmark$	(c) $\frac{x}{e^2} - y - z + 2 = 0 \mathbf{x}$	(d) $1.859 \mathbf{x}$	(e) $16/7 \mathbf{x}$	(f) $\left[\frac{15}{44}, \frac{1}{44}\right] \checkmark$
	$iii : \mathbf{N}$	(a) $\frac{16xy + 7y^2}{4\sqrt[4]{(8x^2y + 7xy^2)^3}} \checkmark$	(b) $\left(\frac{92}{4\sqrt[4]{120^3}}, \frac{88}{4\sqrt[4]{120^3}}\right) \mathbf{x}$	(c) $x + 2z = 0 \checkmark$	(d) $0.005 \mathbf{x}$	(e) $-5/44 \checkmark$	(f) $\left[\frac{-43}{8}, \frac{13}{8}\right] \checkmark$
	$iv : \mathbf{D}$	(a) $\frac{12xy + 8y^2}{2\sqrt{6x^2y + 8xy^2}} \checkmark$	(b) $\left(\frac{128}{2\sqrt[3]{320}}, \frac{224}{2\sqrt[3]{320}}\right) \mathbf{x}$	(c) $\frac{x}{e^2} - y - z + 2 = 0 \mathbf{x}$	(d) $2.071 \mathbf{x}$	(e) $-1/20 \checkmark$	(f) $\left[\frac{85}{46}, \frac{76}{23}\right] \checkmark$
ζ	$i : \mathbf{N}$	(a) $\frac{6x^2 + 10xy}{2\sqrt{6x^2y + 5xy^2}} \checkmark$	(b) $\left(\frac{224}{2\sqrt[4]{456}}, \frac{174}{2\sqrt[4]{456}}\right) \mathbf{x}$	(c) $x - y - z + 0 = 0 \checkmark$	(d) $0.4 \mathbf{x}$	(e) $6/5 \checkmark$	(f) $\left[\frac{-5}{19}, \frac{1}{19}\right] \checkmark$
	$ii : \mathbf{I}$	(a) $\frac{8x^2 + 14xy}{4\sqrt[4]{(8x^2y + 7xy^2)^3}} \mathbf{x}$	(b) $\left(\frac{124}{4\sqrt[4]{228^3}}, \frac{156}{4\sqrt[4]{228^3}}\right) \checkmark$	(c) $x + z = 0 \mathbf{x}$	(d) $-0.1 \mathbf{x}$	(e) $2/5 \mathbf{x}$	(f) $\left[\frac{-7}{23}, \frac{2}{23}\right] \checkmark$
	$iii : \mathbf{N}$	(a) $\frac{2xy + 3y^2}{3\sqrt[3]{(x^2y + 3xy^2)^2}} \checkmark$	(b) $\left(\frac{5}{3\sqrt[3]{4^2}}, \frac{7}{3\sqrt[3]{4^2}}\right) \mathbf{x}$	(c) $x + z = 0 \checkmark$	(d) $0.2 \mathbf{x}$	(e) $27/11 \checkmark$	(f) $\left[\frac{11}{6}, \frac{1}{3}\right] \checkmark$
	$iv : \mathbf{E}$	(a) $\frac{5x^2 + 10xy}{4\sqrt[4]{(5x^2y + 5xy^2)^3}} \checkmark$	(b) $\left(\frac{15}{4\sqrt[4]{10^3}}, \frac{15}{4\sqrt[4]{10^3}}\right) \mathbf{x}$	(c) $x + 2z = 0 \mathbf{x}$	(d) $-0.005 \mathbf{x}$	(e) $2/3 \checkmark$	(f) $\left[\frac{3}{2}, \frac{1}{2}\right] \mathbf{x}$
η	$i : \mathbf{A}$	(a) $\frac{3x^2 + 10xy}{2\sqrt{3x^2y + 5xy^2}} \checkmark$	(b) $\left(\frac{81}{2\sqrt[3]{126}}, \frac{72}{2\sqrt[3]{126}}\right) \mathbf{x}$	(c) $\frac{x}{e^4} - y - z + 4 = 0 \mathbf{x}$	(d) $3.98 \mathbf{x}$	(e) $32/45 \mathbf{x}$	(f) $\left[\frac{101}{30}, \frac{-109}{30}\right] \mathbf{x}$
	$ii : \mathbf{C}$	(a) $\frac{10xy + 3y^2}{3\sqrt[3]{(5x^2y + 3xy^2)^2}} \checkmark$	(b) $\left(\frac{52}{3\sqrt[3]{64^2}}, \frac{44}{3\sqrt[3]{64^2}}\right) \mathbf{x}$	(c) $x - 2z = 0 \mathbf{x}$	(d) $-0.005 \mathbf{x}$	(e) $12/59 \mathbf{x}$	(f) $\left[\frac{-19}{18}, \frac{8}{9}\right] \checkmark$
	$iii : \mathbf{I}$	(a) $\frac{5x^2 + 18xy}{4\sqrt[4]{(5x^2y + 9xy^2)^3}} \mathbf{x}$	(b) $\left(\frac{304}{4\sqrt[4]{896^3}}, \frac{368}{4\sqrt[4]{896^3}}\right) \checkmark$	(c) $x + z = 0 \mathbf{x}$	(d) $0.2 \mathbf{x}$	(e) $-12/11 \mathbf{x}$	(f) $\left[\frac{9}{2}, \frac{-1}{2}\right] \checkmark$
	$iv : \mathbf{D}$	(a) $\frac{12xy + y^2}{4\sqrt[4]{(6x^2y + xy^2)^3}} \checkmark$	(b) $\left(\frac{37}{4\sqrt[4]{57^3}}, \frac{60}{4\sqrt[4]{57^3}}\right) \mathbf{x}$	(c) $\frac{x}{e^2} - y - z + 2 = 0 \mathbf{x}$	(d) $2.341 \mathbf{x}$	(e) $36/11 \checkmark$	(f) $\left[\frac{-29}{18}, \frac{34}{9}\right] \checkmark$

Functions of Two Variables (solutions)

θ	$i : \mathbf{D}$	(a) $\frac{8xy + y^2}{2\sqrt{4x^2y + xy^2}}$ ✓	(b) $\left(\frac{20}{2\sqrt{12}}, \frac{8}{2\sqrt{12}}\right) \mathbf{x}$	(c) $\frac{x}{e^2} - y - z + 2 = 0 \mathbf{x}$	(d) $2.024 \mathbf{x}$	(e) $\frac{1}{18} \mathbf{x}$	(f) $\left[\frac{29}{84}, \frac{-1}{28}\right] \mathbf{x}$
	$ii : \mathbf{O}$	(a) $\frac{2x^2 + 4xy}{2\sqrt{2x^2y + 2xy^2}}$ ✓	(b) $\left(\frac{42}{2\sqrt{60}}, \frac{32}{2\sqrt{60}}\right) \mathbf{x}$	(c) $x + 3z = 0 \mathbf{x}$	(d) $0.067 \mathbf{x}$	(e) $-1/19 \mathbf{x}$	(f) $\left[\frac{-6}{47}, \frac{1}{47}\right] \mathbf{x}$
	$iii : \mathbf{O}$	(a) $\frac{4xy + 3y^2}{2\sqrt{2x^2y + 3xy^2}}$ ✓	(b) $\left(\frac{63}{2\sqrt{135}}, \frac{72}{2\sqrt{135}}\right) \mathbf{x}$	(c) $\frac{x}{e^4} - y - z + 4 = 0 \mathbf{x}$	(d) $3.795 \mathbf{x}$	(e) $\frac{5}{4} \mathbf{x}$	(f) $\left[\frac{7}{4}, \frac{3}{4}\right] \mathbf{x}$
	$iv : \mathbf{R}$	(a) $\frac{2xy + y^2}{3\sqrt{(x^2y + xy^2)^2}}$ ✓	(b) $\left(\frac{3}{3\sqrt[3]{22}}, \frac{3}{3\sqrt[3]{22}}\right) \mathbf{x}$	(c) $\frac{x}{e^3} - y - z + 3 = 0 \mathbf{x}$	(d) $2.9 \mathbf{x}$	(e) $\frac{12}{23} \mathbf{x}$	(f) $\left[\frac{-19}{8}, \frac{1}{4}\right] \mathbf{x}$
ι	$i : \mathbf{F}$	(a) $\frac{5x^2 + 4xy}{3\sqrt{(5x^2y + 2xy^2)^2}}$ ✓	(b) $\left(\frac{22}{3\sqrt[3]{24^2}}, \frac{28}{3\sqrt[3]{24^2}}\right) \mathbf{x}$	(c) $\frac{x}{e} - y - z + 1 = 0 \mathbf{x}$	(d) $1.064 \mathbf{x}$	(e) $2 \mathbf{x}$	(f) $\left[\frac{-1}{3}, \frac{5}{3}\right] \mathbf{x}$
	$ii : \mathbf{A}$	(a) $\frac{12xy + 4y^2}{4\sqrt{(6x^2y + 4xy^2)^3}}$ ✓	(b) $\left(\frac{160}{4\sqrt[4]{224^3}}, \frac{88}{4\sqrt[4]{224^3}}\right) \mathbf{x}$	(c) $x - z = 0 \mathbf{x}$	(d) $0.01 \mathbf{x}$	(e) $\frac{63}{55} \mathbf{x}$	(f) $\left[\frac{57}{56}, \frac{-1}{28}\right] \mathbf{x}$
	$iii : \mathbf{L}$	(a) $\frac{2x^2 + 16xy}{3\sqrt{(2x^2y + 8xy^2)^2}}$ ✓	(b) $\left(\frac{108}{3\sqrt[3]{270^2}}, \frac{162}{3\sqrt[3]{270^2}}\right) \mathbf{x}$	(c) $x + z = 0 \mathbf{x}$	(d) $-0.02 \mathbf{x}$	(e) $-7/136 \mathbf{x}$	(f) $\left[\frac{-61}{74}, \frac{-13}{37}\right] \mathbf{x}$
	$iv : \mathbf{L}$	(a) $\frac{14xy + 5y^2}{4\sqrt[4]{(7x^2y + 5xy^2)^3}}$ ✓	(b) $\left(\frac{192}{4\sqrt[4]{272^3}}, \frac{108}{4\sqrt[4]{272^3}}\right) \mathbf{x}$	(c) $\frac{x}{e^2} - y - z + 2 = 0 \mathbf{x}$	(d) $1.976 \mathbf{x}$	(e) $-36/7 \mathbf{x}$	(f) $[-3, 3] \mathbf{x}$
κ	$i : \mathbf{D}$	(a) $\frac{4xy + 7y^2}{2\sqrt{2x^2y + 7xy^2}}$ ✓	(b) $\left(\frac{75}{2\sqrt{69}}, \frac{44}{2\sqrt{69}}\right) \mathbf{x}$	(c) $\frac{x}{e^4} - y - z + 4 = 0 \mathbf{x}$	(d) $3.698 \mathbf{x}$	(e) $-6/23 \mathbf{x}$	(f) $\left[\frac{-7}{4}, \frac{-1}{2}\right] \mathbf{x}$
	$ii : \mathbf{E}$	(a) $\frac{4x^2 + 18xy}{3\sqrt[3]{(4x^2y + 9xy^2)^2}}$ ✓	(b) $\left(\frac{17}{3\sqrt[3]{13^2}}, \frac{22}{3\sqrt[3]{13^2}}\right) \mathbf{x}$	(c) $x - 3z = 0 \mathbf{x}$	(d) $0.01 \mathbf{x}$	(e) $-3/56 \mathbf{x}$	(f) $\left[\frac{31}{20}, \frac{-9}{10}\right] \mathbf{x}$
	$iii : \mathbf{S}$	(a) $\frac{12xy + 5y^2}{2\sqrt{6x^2y + 5xy^2}}$ $\mathbf{x}$	(b) $\left(\frac{68}{2\sqrt{88}}, \frac{64}{2\sqrt{88}}\right) \mathbf{x}$	(c) $x - y - z + 0 = 0 \mathbf{x}$	(d) $0.0 \mathbf{x}$	(e) $\frac{4}{3} \mathbf{x}$	(f) $\left[\frac{19}{3}, \frac{1}{3}\right] \mathbf{x}$
	$iv : \mathbf{K}$	(a) $\frac{5x^2 + 2xy}{3\sqrt[3]{(5x^2y + xy^2)^2}}$ ✓	(b) $\left(\frac{176}{3\sqrt[3]{384^2}}, \frac{112}{3\sqrt[3]{384^2}}\right) \mathbf{x}$	(c) $x + z = 0 \mathbf{x}$	(d) $-0.2 \mathbf{x}$	(e) $\frac{9}{2} \mathbf{x}$	(f) $\left[\frac{-9}{4}, \frac{1}{4}\right] \mathbf{x}$
λ	$i : \mathbf{T}$	(a) $\frac{4x^2 + 8xy}{4\sqrt[4]{(4x^2y + 4xy^2)^3}}$ $\mathbf{x}$	(b) $\left(\frac{192}{4\sqrt[4]{512^3}}, \frac{192}{4\sqrt[4]{512^3}}\right) \mathbf{x}$	(c) $x + 3z = 0 \mathbf{x}$	(d) $0.007 \mathbf{x}$	(e) $\frac{4}{15} \mathbf{x}$	(f) $\left[\frac{7}{15}, \frac{-8}{15}\right] \mathbf{x}$
	$ii : \mathbf{O}$	(a) $\frac{x^2 + 6xy}{4\sqrt[4]{(x^2y + 3xy^2)^3}}$ ✓	(b) $\left(\frac{7}{4\sqrt[4]{10^3}}, \frac{16}{4\sqrt[4]{10^3}}\right) \mathbf{x}$	(c) $\frac{x}{e^4} - y - z + 4 = 0 \mathbf{x}$	(d) $4.03 \mathbf{x}$	(e) $-9/8 \mathbf{x}$	(f) $\left[\frac{9}{22}, \frac{13}{22}\right] \mathbf{x}$
	$iii : \mathbf{W}$	(a) $\frac{x^2 + 2xy}{2\sqrt{x^2y + xy^2}}$ ✓	(b) $\left(\frac{21}{2\sqrt{30}}, \frac{16}{2\sqrt{30}}\right) \mathbf{x}$	(c) $\frac{x}{e^4} - y - z + 4 = 0 \mathbf{x}$	(d) $3.99 \mathbf{x}$	(e) $-5/51 \mathbf{x}$	(f) $\left[\frac{-3}{2}, \frac{1}{2}\right] \mathbf{x}$
	$iv : \mathbf{N}$	(a) $\frac{6xy + y^2}{2\sqrt{3x^2y + xy^2}}$ ✓	(b) $\left(\frac{28}{2\sqrt{32}}, \frac{20}{2\sqrt{32}}\right) \mathbf{x}$	(c) $x + 3z = 0 \mathbf{x}$	(d) $-0.033 \mathbf{x}$	(e) $\frac{3}{35} \mathbf{x}$	(f) $\left[\frac{5}{13}, \frac{3}{13}\right] \mathbf{x}$
μ	$i : \mathbf{T}$	(a) $\frac{6x^2 + 8xy}{2\sqrt{6x^2y + 4xy^2}}$ $\mathbf{x}$	(b) $\left(\frac{64}{2\sqrt{80}}, \frac{56}{2\sqrt{80}}\right) \mathbf{x}$	(c) $x - y - z + 0 = 0 \mathbf{x}$	(d) $-0.07 \mathbf{x}$	(e) $1 \mathbf{x}$	(f) $\left[\frac{2}{13}, \frac{-5}{13}\right] \mathbf{x}$
	$ii : \mathbf{U}$	(a) $\frac{2x^2 + 2xy}{2\sqrt{2x^2y + xy^2}}$ ✓	(b) $\left(\frac{13}{2\sqrt{21}}, \frac{24}{2\sqrt{21}}\right) \mathbf{x}$	(c) $x + 2z = 0 \mathbf{x}$	(d) $-0.1 \mathbf{x}$	(e) $-49/12 \mathbf{x}$	(f) $\left[\frac{-19}{24}, \frac{-29}{24}\right] \mathbf{x}$
	$iii : \mathbf{R}$	(a) $\frac{x^2 + 14xy}{2\sqrt{x^2y + 7xy^2}}$ ✓	(b) $\left(\frac{44}{2\sqrt{144}}, \frac{128}{2\sqrt{144}}\right) \mathbf{x}$	(c) $x - z = 0 \mathbf{x}$	(d) $0.2 \mathbf{x}$	(e) $\frac{27}{23} \mathbf{x}$	(f) $\left[\frac{-3}{2}, 0\right] \mathbf{x}$
	$iv : \mathbf{N}$	(a) $\frac{3x^2 + 14xy}{3\sqrt[3]{(3x^2y + 7xy^2)^2}}$ ✓	(b) $\left(\frac{52}{3\sqrt[3]{80^2}}, \frac{68}{3\sqrt[3]{80^2}}\right) \mathbf{x}$	(c) $x + 3z = 0 \mathbf{x}$	(d) $-0.003 \mathbf{x}$	(e) $\frac{1}{4} \mathbf{x}$	(f) $\left[\frac{-41}{52}, \frac{19}{52}\right] \mathbf{x}$
ν	$i : \mathbf{D}$	(a) $\frac{2xy + 9y^2}{3\sqrt[3]{(x^2y + 9xy^2)^2}}$ ✓	(b) $\left(\frac{176}{3\sqrt[3]{640^2}}, \frac{304}{3\sqrt[3]{640^2}}\right) \mathbf{x}$	(c) $\frac{x}{e^3} - y - z + 3 = 0 \mathbf{x}$	(d) $2.995 \mathbf{x}$	(e) $-49/16 \mathbf{x}$	(f) $\left[\frac{1}{2}, 0\right] \mathbf{x}$
	$ii : \mathbf{I}$	(a) $\frac{8x^2 + 6xy}{3\sqrt[3]{(8x^2y + 3xy^2)^2}}$ $\mathbf{x}$	(b) $\left(\frac{76}{3\sqrt[3]{88^2}}, \frac{56}{3\sqrt[3]{88^2}}\right) \mathbf{x}$	(c) $\frac{x}{e} - y - z + 1 = 0 \mathbf{x}$	(d) $1.084 \mathbf{x}$	(e) $-6/95 \mathbf{x}$	(f) $\left[\frac{7}{2}, \frac{-5}{2}\right] \mathbf{x}$
	$iii : \mathbf{R}$	(a) $\frac{2xy + 2y^2}{3\sqrt[3]{(x^2y + 2xy^2)^2}}$ ✓	(b) $\left(\frac{6}{3\sqrt[3]{8^2}}, \frac{12}{3\sqrt[3]{8^2}}\right) \mathbf{x}$	(c) $x + z = 0 \mathbf{x}$	(d) $-0.01 \mathbf{x}$	(e) $-1/3 \mathbf{x}$	(f) $[1, 1] \mathbf{x}$
	$iv : \mathbf{T}$	(a) $\frac{16xy + 8y^2}{2\sqrt{8x^2y + 8xy^2}}$ $\mathbf{x}$	(b) $\left(\frac{384}{2\sqrt{1024}}, \frac{384}{2\sqrt{1024}}\right) \mathbf{x}$	(c) $x - z = 0 \mathbf{x}$	(d) $0.1 \mathbf{x}$	(e) $-25/3 \mathbf{x}$	(f) $\left[\frac{41}{42}, \frac{1}{21}\right] \mathbf{x}$
ξ	$i : \mathbf{F}$	(a) $\frac{7x^2 + 6xy}{3\sqrt[3]{(7x^2y + 3xy^2)^2}}$ ✓	(b) $\left(\frac{124}{3\sqrt[3]{272^2}}, \frac{160}{3\sqrt[3]{272^2}}\right) \mathbf{x}$	(c) $x - 3z = 0 \mathbf{x}$	(d) $-0.01 \mathbf{x}$	(e) $\frac{16}{7} \mathbf{x}$	(f) $\left[\frac{26}{11}, \frac{-14}{11}\right] \mathbf{x}$
	$ii : \mathbf{I}$	(a) $\frac{16xy + 3y^2}{3\sqrt[3]{(8x^2y + 3xy^2)^2}}$ $\mathbf{x}$	(b) $\left(\frac{108}{3\sqrt[3]{180^2}}, \frac{108}{3\sqrt[3]{180^2}}\right) \mathbf{x}$	(c) $x + 2z = 0 \mathbf{x}$	(d) $-0.01 \mathbf{x}$	(e) $\frac{50}{23} \mathbf{x}$	(f) $\left[\frac{-20}{9}, \frac{25}{9}\right] \mathbf{x}$
	$iii : \mathbf{V}$	(a) $\frac{3x^2 + 4xy}{2\sqrt{3x^2y + 2xy^2}}$ ✓	(b) $\left(\frac{26}{2\sqrt{56}}, \frac{64}{2\sqrt{56}}\right) \mathbf{x}$	(c) $\frac{x}{e^3} - y - z + 3 = 0 \mathbf{x}$	(d) $3.025 \mathbf{x}$	(e) $\frac{5}{71} \mathbf{x}$	(f) $\left[\frac{11}{12}, \frac{-13}{12}\right] \mathbf{x}$
	$iv : \mathbf{E}$	(a) $\frac{2x^2 + 2xy}{3\sqrt[3]{(2x^2y + xy^2)^2}}$ ✓	(b) $\left(\frac{32}{3\sqrt[3]{24^2}}, \frac{10}{3\sqrt[3]{24^2}}\right) \mathbf{x}$	(c) $\frac{x}{e^4} - y - z + 4 = 0 \mathbf{x}$	(d) $3.966 \mathbf{x}$	(e) $\frac{98}{15} \mathbf{x}$	(f) $\left[\frac{15}{68}, \frac{-7}{68}\right] \mathbf{x}$

Functions of Two Variables (solutions)

$\square$	$i : \mathbf{G}$	(a) $\frac{3x^2 + 12xy}{4\sqrt[4]{(3x^2y + 6xy^2)^3}}$ ✓	(b) $\left(\frac{24}{4\sqrt[4]{45^3}}, \frac{63}{4\sqrt[4]{45^3}}\right)$ ✓	(c) $x + 3z = 0 \mathbf{x}$	(d) $0.007 \mathbf{x}$	(e) $\frac{98}{23} \checkmark$	(f) $\left[\frac{5}{47}, \frac{12}{47}\right]$ ✓
	$ii : \mathbf{I}$	(a) $\frac{6x^2 + 8xy}{2\sqrt[4]{6x^2y + 4xy^2}} \mathbf{x}$	(b) $\left(\frac{52}{2\sqrt[4]{112}}, \frac{128}{2\sqrt[4]{112}}\right)$ ✓	(c) $x - z = 0 \mathbf{x}$	(d) $0.1 \mathbf{x}$	(e) $\frac{8}{7} \mathbf{x}$	(f) $[1, -3]$ ✓
	$iii : \mathbf{F}$	(a) $\frac{8xy + 4y^2}{2\sqrt[4]{4x^2y + 4xy^2}}$ ✓	(b) $\left(\frac{128}{2\sqrt[4]{192}}, \frac{80}{2\sqrt[4]{192}}\right)$ ✓	(c) $\frac{x}{e^4} - y - z + 4 = 0 \mathbf{x}$	(d) $3.905 \mathbf{x}$	(e) $-1/_{15} \mathbf{x}$	(f) $\left[\frac{-53}{48}, \frac{19}{12}\right]$ ✓
	$iv : \mathbf{T}$	(a) $\frac{9x^2 + 2xy}{2\sqrt[4]{9x^2y + xy^2}} \mathbf{x}$	(b) $\left(\frac{40}{2\sqrt[4]{22}}, \frac{13}{2\sqrt[4]{22}}\right)$ ✓	(c) $x + z = 0 \checkmark$	(d) $-0.02 \mathbf{x}$	(e) $\frac{36}{47} \checkmark$	(f) $\left[\frac{-57}{34}, \frac{11}{17}\right]$ ✓
π	$i : \mathbf{S}$	(a) $\frac{6xy + 6y^2}{4\sqrt[4]{(3x^2y + 6xy^2)^3}} \mathbf{x}$	(b) $\left(\frac{90}{4\sqrt[4]{144^3}}, \frac{84}{4\sqrt[4]{144^3}}\right)$ ✓	(c) $x - y - z + 0 = 0 \checkmark$	(d) $-0.04 \mathbf{x}$	(e) $-25/_{11} \mathbf{x}$	(f) $\left[\frac{3}{10}, \frac{11}{10}\right]$ ✓
	$ii : \mathbf{T}$	(a) $\frac{8xy + 6y^2}{2\sqrt[4]{4x^2y + 6xy^2}} \mathbf{x}$	(b) $\left(\frac{160}{2\sqrt[4]{256}}, \frac{112}{2\sqrt[4]{256}}\right)$ ✓	(c) $x + 2z = 0 \checkmark$	(d) $-0.015 \mathbf{x}$	(e) $-9/_{11} \checkmark$	(f) $\left[\frac{-23}{18}, \frac{13}{9}\right]$ ✓
	$iii : \mathbf{A}$	(a) $\frac{4xy + 3y^2}{3\sqrt[3]{(2x^2y + 3xy^2)^2}} \checkmark$	(b) $\left(\frac{96}{3\sqrt[3]{216^2}}, \frac{90}{3\sqrt[3]{216^2}}\right) \mathbf{x}$	(c) $x + z = 0 \mathbf{x}$	(d) $0.2 \mathbf{x}$	(e) $\frac{9}{4} \mathbf{x}$	(f) $\left[\frac{69}{98}, \frac{-87}{98}\right] \mathbf{x}$
	$iv : \mathbf{R}$	(a) $\frac{14xy + 6y^2}{2\sqrt[4]{7x^2y + 6xy^2}} \checkmark$	(b) $\left(\frac{108}{2\sqrt[4]{198}}, \frac{135}{2\sqrt[4]{198}}\right)$ ✓	(c) $x + 3z = 0 \checkmark$	(d) $0.007 \mathbf{x}$	(e) $-3/_{11} \checkmark$	(f) $\left[\frac{-5}{27}, \frac{7}{27}\right] \mathbf{x}$
ρ	$i : \mathbf{S}$	(a) $\frac{6xy + 7y^2}{2\sqrt[4]{3x^2y + 7xy^2}} \mathbf{x}$	(b) $\left(\frac{136}{2\sqrt[4]{124}}, \frac{59}{2\sqrt[4]{124}}\right)$ ✓	(c) $x + z = 0 \checkmark$	(d) $0.1 \mathbf{x}$	(e) $-9/_{23} \mathbf{x}$	(f) $\left[\frac{-3}{4}, \frac{9}{4}\right]$ ✓
	$ii : \mathbf{H}$	(a) $\frac{18xy + 3y^2}{3\sqrt[3]{(9x^2y + 3xy^2)^2}} \checkmark$	(b) $\left(\frac{21}{3\sqrt[3]{12^2}}, \frac{15}{3\sqrt[3]{12^2}}\right)$ ✓	(c) $x - y - z + 0 = 0 \mathbf{x}$	(d) $0.08 \mathbf{x}$	(e) $-20/_{11} \checkmark$	(f) $\left[\frac{-71}{54}, \frac{10}{27}\right] \mathbf{x}$
	$iii : \mathbf{O}$	(a) $\frac{8xy + 7y^2}{2\sqrt[4]{4x^2y + 7xy^2}} \checkmark$	(b) $\left(\frac{144}{2\sqrt[4]{128}}, \frac{60}{2\sqrt[4]{128}}\right) \mathbf{x}$	(c) $x - y - z + 0 = 0 \checkmark$	(d) $-0.01 \mathbf{x}$	(e) $\frac{28}{171} \checkmark$	(f) $\left[\frac{101}{90}, \frac{-34}{45}\right] \mathbf{x}$
	$iv : \mathbf{P}$	(a) $\frac{3x^2 + 2xy}{2\sqrt[4]{3x^2y + xy^2}} \checkmark$	(b) $\left(\frac{16}{2\sqrt[4]{10}}, \frac{7}{2\sqrt[4]{10}}\right)$ ✓	(c) $x + 3z = 0 \checkmark$	(d) $0.007 \mathbf{x}$	(e) $\frac{98}{47} \mathbf{x}$	(f) $\left[\frac{2}{17}, \frac{7}{17}\right]$ ✓
σ	$i : \mathbf{W}$	(a) $\frac{2x^2 + 10xy}{2\sqrt[4]{2x^2y + 5xy^2}} \checkmark$	(b) $\left(\frac{52}{2\sqrt[4]{14}}, \frac{112}{2\sqrt[4]{14}}\right)$ ✓	(c) $x - z = 0 \checkmark$	(d) $-0.2 \checkmark$	(e) $-8/_{23} \checkmark$	(f) $\left[\frac{3}{2}, \frac{-11}{2}\right] \mathbf{x}$
	$ii : \mathbf{O}$	(a) $\frac{2x^2 + 8xy}{4\sqrt[4]{(2x^2y + 4xy^2)^3}} \checkmark$	(b) $\left(\frac{16}{4\sqrt[4]{30^3}}, \frac{42}{4\sqrt[4]{30^3}}\right) \mathbf{x}$	(c) $x + 2z = 0 \checkmark$	(d) $0.05 \mathbf{x}$	(e) $-1/_{19} \checkmark$	(f) $\left[\frac{23}{16}, \frac{25}{16}\right] \mathbf{x}$
	$iii : \mathbf{O}$	(a) $\frac{8x^2 + 10xy}{3\sqrt[3]{(8x^2y + 5xy^2)^2}} \checkmark$	(b) $\left(\frac{141}{3\sqrt[3]{186^2}}, \frac{92}{3\sqrt[3]{186^2}}\right) \mathbf{x}$	(c) $\frac{x}{e^4} - y - z + 4 = 0 \checkmark$	(d) $4.201 \mathbf{x}$	(e) $-28/_{103} \checkmark$	(f) $\left[\frac{-43}{152}, \frac{-23}{152}\right] \mathbf{x}$
	$iv : \mathbf{D}$	(a) $\frac{2xy + 2y^2}{4\sqrt[4]{(x^2y + 2xy^2)^3}} \checkmark$	(b) $\left(\frac{16}{4\sqrt[4]{24^3}}, \frac{20}{4\sqrt[4]{24^3}}\right) \mathbf{x}$	(c) $x + z = 0 \mathbf{x}$	(d) $0.01 \mathbf{x}$	(e) $-1/_{21} \checkmark$	(f) $\left[\frac{-29}{48}, \frac{-3}{16}\right]$ ✓
τ	$i : \mathbf{C}$	(a) $\frac{3x^2 + 4xy}{2\sqrt[4]{3x^2y + 2xy^2}} \checkmark$	(b) $\left(\frac{104}{2\sqrt[4]{204}}, \frac{75}{2\sqrt[4]{204}}\right) \mathbf{x}$	(c) $\frac{x}{e^4} - y - z + 4 = 0 \mathbf{x}$	(d) $4.031 \mathbf{x}$	(e) $\frac{9}{8} \mathbf{x}$	(f) $\left[\frac{-33}{166}, \frac{17}{83}\right]$ ✓
	$ii : \mathbf{E}$	(a) $\frac{4xy + 6y^2}{4\sqrt[4]{(2x^2y + 6xy^2)^3}} \checkmark$	(b) $\left(\frac{102}{4\sqrt[4]{312^3}}, \frac{176}{4\sqrt[4]{312^3}}\right) \mathbf{x}$	(c) $x - z = 0 \mathbf{x}$	(d) $-0.02 \mathbf{x}$	(e) $-7/_{3} \checkmark$	(f) $\left[\frac{-31}{37}, \frac{12}{37}\right] \mathbf{x}$
	$iii : \mathbf{L}$	(a) $\frac{6xy + 7y^2}{3\sqrt[3]{(3x^2y + 7xy^2)^2}} \checkmark$	(b) $\left(\frac{208}{3\sqrt[3]{640^2}}, \frac{272}{3\sqrt[3]{640^2}}\right)$ ✓	(c) $x + z = 0 \checkmark$	(d) $0.1 \mathbf{x}$	(e) $-9/_{8} \mathbf{x}$	(f) $\left[\frac{-3}{4}, \frac{-21}{4}\right] \mathbf{x}$
	$iv : \mathbf{L}$	(a) $\frac{4xy + 2y^2}{3\sqrt[3]{(2x^2y + 2xy^2)^2}} \checkmark$	(b) $\left(\frac{96}{3\sqrt[3]{256^2}}, \frac{96}{3\sqrt[3]{256^2}}\right)$ ✓	(c) $x + 2z = 0 \checkmark$	(d) $-0.1 \mathbf{x}$	(e) $\frac{7}{26} \mathbf{x}$	(f) $\left[\frac{16}{45}, \frac{-13}{45}\right] \mathbf{x}$
v	$i : \mathbf{F}$	(a) $\frac{x^2 + 12xy}{2\sqrt[4]{x^2y + 6xy^2}} \checkmark$	(b) $\left(\frac{10}{2\sqrt[4]{16}}, \frac{28}{2\sqrt[4]{16}}\right)$ ✓	(c) $x + 2z = 0 \mathbf{x}$	(d) $-0.005 \mathbf{x}$	(e) $-8/_{7} \mathbf{x}$	(f) $[2, 0]$ ✓
	$ii : \mathbf{O}$	(a) $\frac{6x^2 + 4xy}{4\sqrt[4]{(6x^2y + 2xy^2)^3}} \checkmark$	(b) $\left(\frac{128}{4\sqrt[4]{160^3}}, \frac{56}{4\sqrt[4]{160^3}}\right) \mathbf{x}$	(c) $\frac{x}{e^4} - y - z + 4 = 0 \checkmark$	(d) $4.014 \mathbf{x}$	(e) $-20/_{39} \checkmark$	(f) $\left[\frac{-17}{40}, \frac{-3}{20}\right] \mathbf{x}$
	$iii : \mathbf{O}$	(a) $\frac{5x^2 + 4xy}{4\sqrt[4]{(5x^2y + 2xy^2)^3}} \checkmark$	(b) $\left(\frac{42}{4\sqrt[4]{88^3}}, \frac{96}{4\sqrt[4]{88^3}}\right) \mathbf{x}$	(c) $\frac{x}{e^2} - y - z + 2 = 0 \checkmark$	(d) $1.979 \mathbf{x}$	(e) $-12/_{35} \checkmark$	(f) $\left[\frac{-1}{2}, 0\right] \mathbf{x}$
	$iv : \mathbf{D}$	(a) $\frac{7x^2 + 10xy}{2\sqrt[4]{7x^2y + 5xy^2}} \checkmark$	(b) $\left(\frac{33}{2\sqrt[4]{38}}, \frac{48}{2\sqrt[4]{38}}\right) \mathbf{x}$	(c) $x - y - z + 0 = 0 \mathbf{x}$	(d) $0.33 \mathbf{x}$	(e) $-24/_{41} \checkmark$	(f) $\left[\frac{-37}{11}, \frac{40}{11}\right]$ ✓
ϕ	$i : \mathbf{B}$	(a) $\frac{8x^2 + 6xy}{4\sqrt[4]{(8x^2y + 3xy^2)^3}} \checkmark$	(b) $\left(\frac{44}{4\sqrt[4]{28^3}}, \frac{20}{4\sqrt[4]{28^3}}\right)$ ✓	(c) $\frac{x}{e^2} - y - z + 2 = 0 \mathbf{x}$	(d) $1.943 \mathbf{x}$	(e) $\frac{25}{16} \mathbf{x}$	(f) $\left[\frac{-5}{8}, \frac{-59}{8}\right] \mathbf{x}$
	$ii : \mathbf{O}$	(a) $\frac{8xy + 9y^2}{2\sqrt[4]{4x^2y + 9xy^2}} \checkmark$	(b) $\left(\frac{25}{2\sqrt[4]{34}}, \frac{52}{2\sqrt[4]{34}}\right) \mathbf{x}$	(c) $x + z = 0 \checkmark$	(d) $0.1 \mathbf{x}$	(e) $-16/_{3} \checkmark$	(f) $\left[\frac{-1}{2}, -3\right] \mathbf{x}$
	$iii : \mathbf{D}$	(a) $\frac{4xy + 2y^2}{3\sqrt[3]{(2x^2y + 2xy^2)^2}} \checkmark$	(b) $\left(\frac{64}{3\sqrt[3]{96^2}}, \frac{40}{3\sqrt[3]{96^2}}\right) \mathbf{x}$	(c) $\frac{x}{e} - y - z + 1 = 0 \mathbf{x}$	(d) $1.054 \mathbf{x}$	(e) $-8/_{13} \checkmark$	(f) $\left[\frac{1}{4}, 0\right]$ ✓
	$iv : \mathbf{Y}$	(a) $\frac{6xy + 7y^2}{4\sqrt[4]{(3x^2y + 7xy^2)^3}} \checkmark$	(b) $\left(\frac{160}{4\sqrt[4]{272^3}}, \frac{124}{4\sqrt[4]{272^3}}\right) \mathbf{x}$	(c) $x + 2z = 0 \checkmark$	(d) $-0.01 \checkmark$	(e) $\frac{75}{56} \checkmark$	(f) $\left[\frac{-9}{10}, \frac{11}{10}\right]$ ✓

Functions of Two Variables (solutions)

$\mathbf{x}$	$i : \mathbf{W}$	(a) $\frac{x^2 + 6xy}{2\sqrt{x^2y + 3xy^2}}$ ✓	(b) $\left(\frac{45}{2\sqrt{108}}, \frac{63}{2\sqrt{108}}\right)$ ✓	(c) $x - 2z = 0$ ✓	(d) 0.1 ✓	(e) $-1/16$ ✓	(f) $\left[\frac{1}{8}, \frac{-47}{8}\right]$ ✗
	$ii : \mathbf{O}$	(a) $\frac{4x^2 + 16xy}{3\sqrt[3]{(4x^2y + 8xy^2)^2}}$ ✓	(b) $\left(\frac{40}{3\sqrt[3]{96^2}}, \frac{128}{3\sqrt[3]{96^2}}\right)$ ✗	(c) $\frac{x}{e^2} - y - z + 2 = 0$ ✓	(d) 1.974 ✗	(e) -5 ✓	(f) $\left[\frac{-18}{5}, \frac{7}{5}\right]$ ✗
	$iii : \mathbf{R}$	(a) $\frac{14xy + 7y^2}{4\sqrt[4]{(7x^2y + 7xy^2)^3}}$ ✓	(b) $\left(\frac{147}{4\sqrt[4]{210^3}}, \frac{112}{4\sqrt[4]{210^3}}\right)$ ✓	(c) $\frac{x}{e^5} - y - z + 5 = 0$ ✓	(d) 4.7 ✗	(e) $5/71$ ✓	(f) $\left[\frac{1}{2}, \frac{-1}{2}\right]$ ✗
	$iv : \mathbf{K}$	(a) $\frac{x^2 + 6xy}{2\sqrt{x^2y + 3xy^2}}$ ✓	(b) $\left(\frac{33}{2\sqrt{30}}, \frac{19}{2\sqrt{30}}\right)$ ✗	(c) $x + z = 0$ ✓	(d) -0.1 ✗	(e) $80/11$ ✗	(f) $\left[\frac{-1}{2}, \frac{1}{2}\right]$ ✗
ψ	$i : \mathbf{F}$	(a) $\frac{2xy + 3y^2}{2\sqrt{x^2y + 3xy^2}}$ ✓	(b) $\left(\frac{20}{2\sqrt{32}}, \frac{28}{2\sqrt{32}}\right)$ ✓	(c) $x + 2z = 0$ ✗	(d) -0.15 ✗	(e) $-25/8$ ✗	(f) $\left[\frac{-31}{6}, \frac{-23}{6}\right]$ ✓
	$ii : \mathbf{I}$	(a) $\frac{5x^2 + 8xy}{3\sqrt[3]{(5x^2y + 4xy^2)^2}}$ ✗	(b) $\left(\frac{36}{3\sqrt[3]{26^2}}, \frac{21}{3\sqrt[3]{26^2}}\right)$ ✓	(c) $\frac{x}{e^4} - y - z + 4 = 0$ ✗	(d) 4.011 ✗	(e) $-36/71$ ✗	(f) $\left[\frac{-9}{5}, \frac{-16}{5}\right]$ ✓
	$iii : \mathbf{S}$	(a) $\frac{x^2 + 10xy}{3\sqrt[3]{(x^2y + 5xy^2)^2}}$ ✗	(b) $\left(\frac{96}{3\sqrt[3]{176^2}}, \frac{84}{3\sqrt[3]{176^2}}\right)$ ✓	(c) $\frac{x}{e} - y - z + 1 = 0$ ✓	(d) 1.274 ✗	(e) $12/17$ ✗	(f) $\left[\frac{-32}{39}, \frac{-7}{13}\right]$ ✓
	$iv : \mathbf{H}$	(a) $\frac{2x^2 + 2xy}{2\sqrt{2x^2y + xy^2}}$ ✓	(b) $\left(\frac{17}{2\sqrt{36}}, \frac{40}{2\sqrt{36}}\right)$ ✓	(c) $x + 3z = 0$ ✗	(d) -0.003 ✗	(e) $-81/16$ ✓	(f) $\left[\frac{21}{80}, \frac{-5}{16}\right]$ ✗
ω	$i : \mathbf{L}$	(a) $\frac{16xy + 3y^2}{2\sqrt{8x^2y + 3xy^2}}$ ✓	(b) $\left(\frac{112}{2\sqrt{80}}, \frac{32}{2\sqrt{80}}\right)$ ✓	(c) $\frac{x}{e^2} - y - z + 2 = 0$ ✓	(d) 2.314 ✗	(e) $2/39$ ✗	(f) $\left[\frac{9}{50}, \frac{1}{10}\right]$ ✗
	$ii : \mathbf{E}$	(a) $\frac{5x^2 + 2xy}{4\sqrt[4]{(5x^2y + xy^2)^3}}$ ✓	(b) $\left(\frac{41}{4\sqrt[4]{84^3}}, \frac{88}{4\sqrt[4]{84^3}}\right)$ ✗	(c) $x + 2z = 0$ ✗	(d) -0.005 ✗	(e) $3/11$ ✓	(f) $\left[\frac{32}{15}, \frac{11}{15}\right]$ ✗
	$iii : \mathbf{A}$	(a) $\frac{8x^2 + 2xy}{4\sqrt[4]{(8x^2y + xy^2)^3}}$ ✓	(b) $\left(\frac{272}{4\sqrt[4]{576^3}}, \frac{160}{4\sqrt[4]{576^3}}\right)$ ✗	(c) $\frac{x}{e^5} - y - z + 5 = 0$ ✗	(d) 4.981 ✗	(e) $98/31$ ✗	(f) $\left[\frac{21}{10}, \frac{-49}{10}\right]$ ✗
	$iv : \mathbf{D}$	(a) $\frac{2xy + y^2}{2\sqrt{x^2y + xy^2}}$ ✓	(b) $\left(\frac{15}{2\sqrt{12}}, \frac{7}{2\sqrt{12}}\right)$ ✗	(c) $\frac{x}{e^2} - y - z + 2 = 0$ ✗	(d) 1.714 ✗	(e) $-36/23$ ✓	(f) $\left[\frac{-25}{18}, \frac{-7}{9}\right]$ ✓